

PROJECT INFORMATION	
STRUCTURE TYPE:	SELF SUPPORTING TOWER (195'-4")
LATITUDE:	40°17'21.44" N / 40.289289°
LONGITUDE:	98°15'55.07" W / -98.265297°
ADDRESS:	150 S. SMITH STREET
CITY, STATE, ZIP:	LAWRENCE, NE 88957
COUNTY:	NUCKOLLS
MARKET:	NEBRASKA
JURISDICTION:	LAWRENCE CITY
SITE MANAGER:	INDUSTRIAL TOWER WEST 2825 EAST BEAVER AVENUE, FORT MORGAN, CO 80701.
CONTACT INFORMATION:	SKYLYN SEIMER - 970-768-0040 JOE DILKS - 970-467-3744

CODE COMPLIANCE	
ALL WORK AND MATERIALS SHALL BE PERFORMED AND INSTALLED IN ACCORDANCE WITH THE CURRENT EDITIONS OF THE FOLLOWING CODES AS ADOPTED BY THE LOCAL GOVERNING AUTHORITIES. NOTHING IN THESE PLANS IS TO BE CONSTRUED TO PERMIT WORK NOT CONFORMING TO THESE CODES	
1. INTERNATIONAL BUILDING CODE (2018) 2. NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) 70-14, NATIONAL ELECTRICAL CODE (NEC) 2017 EDITION 3. ASCE 7-16 4. TIA-222-H 5. LIFE SAFETY CODE NFPA-101-2021 6. AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC) 360-10 AND 341-10 7. UNDERWRITERS LABORATORIES (U L) APPROVED ELECTRICAL PRODUCTS 8. LOCAL JURISDICTIONAL REQUIREMENTS 9. CITY/COUNTY ORDINANCES 10. INTERNATIONAL PLUMBING CODE (2018) 11. INTERNATIONAL FUEL GAS ADOPTION AND AMENDMENTS (2015)	

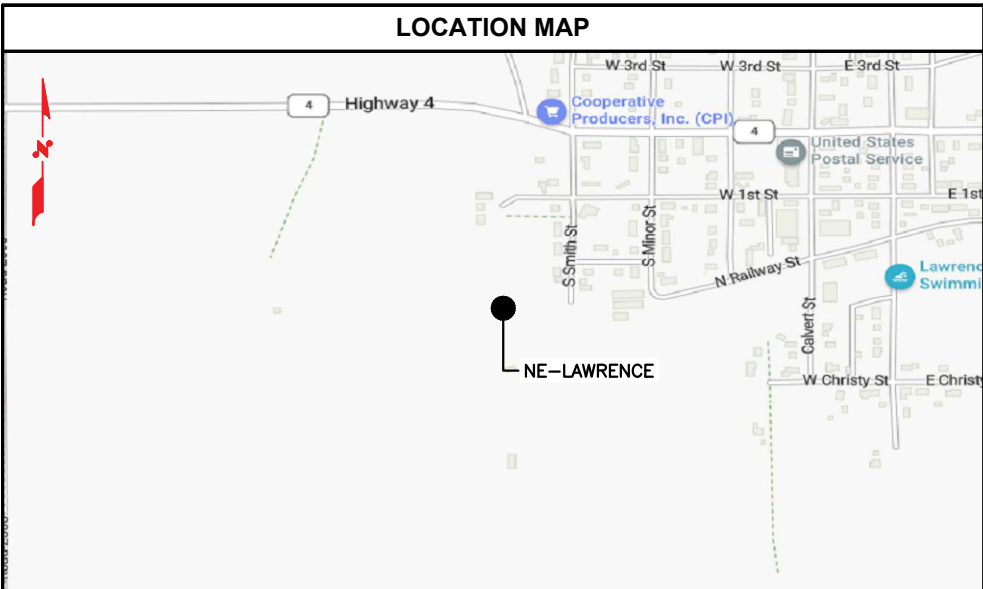
APPROVALS	
PROJECT MANAGER	DATE
CONSTRUCTION	DATE
RF ENGINEERING	DATE
ZONING / SITE ACQ.	DATE
OPERATIONS	DATE
TOWER OWNER	DATE

DRAWING INDEX		
SHEET NO.	DESCRIPTION	REV.
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A-2	EXISTING EQUIPMENT SHELTER DETAILS	0
A-3	PROPOSED EQUIPMENT SHELTER DETAILS	0
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A-5	EXISTING MW ANTENNA PLAN VIEW ELEV: 150'-5", 123'-7"	0
A-6	PROPOSED ANTENNA PLAN VIEW (TEMPORARY) (HOT CUT)	0
A-7	PROPOSED ANTENNA PLAN VIEW (FINAL) ELEV: 192'-4"	0
A-8	PROPOSED MW ANTENNA PLAN ELEV	0
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A-15	RF EQUIPMENT DETAILS	0
A-16	MW EQUIPMENT DETAILS	0
GR-1	GROUNDING DETAILS	0
ADDENDUM - MOUNT DETAILS		-

SITE NAME: NE-LAWRENCE

SITE ID: 752
150 S. SMITH STREET
LAWRENCE, NE 88957

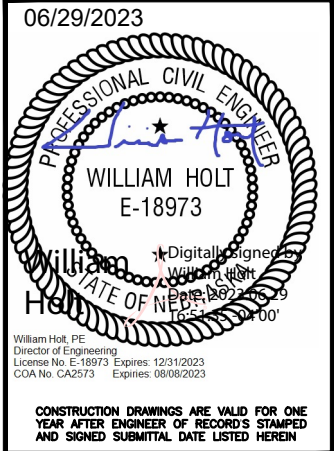
PROJECT NAME RIP & REPLACE



DRIVING DIRECTIONS	
CENTRAL NEBRASKA REGIONAL AIRPORT: HEAD WEST ON AIRPORT SERVICE RD TOWARD BARON LN, TURN RIGHT ONTO BARON LN, TURN LEFT ONTO SKY PARK RD, TURN RIGHT ONTO E AIRPORT RD / COUNTY HWY-37, KEEP STRAIGHT TO GET ONTO US-281 S / N US HIGHWAY 281, TURN LEFT ONTO W 42ND ST, KEEP STRAIGHT TO GET ONTO E 42ND ST, KEEP STRAIGHT TO GET ONTO N SHOWBOAT BLVD, KEEP STRAIGHT TO GET ONTO S SHOWBOAT BLVD, TURN LEFT TO STAY ON S SHOWBOAT BLVD, KEEP LEFT TO STAY ON S SHOWBOAT BLVD, KEEP STRAIGHT TO GET ONTO E SUNDOWN RD, BEAR RIGHT ONTO S SHOWBOAT BLVD, KEEP STRAIGHT TO GET ONTO E BLUE VALLEY RD, KEEP STRAIGHT TO GET ONTO NE-78 / HIGHWAY 78, TURN LEFT ONTO NE-4 / HIGHWAY 4, TURN RIGHT ONTO N SMITH ST, ROAD NAME CHANGES TO S SMITH ST, ARRIVE AT S SMITH ST ON THE RIGHT.	

GENERAL NOTES	
1. THIS DOCUMENT IS THE CREATION, DESIGN, PROPERTY AND COPYRIGHTED WORK OF VIAERO. ANY DUPLICATION OR USE WITHOUT EXPRESS WRITTEN CONSENT IS STRICTLY PROHIBITED. DUPLICATION AND USE BY GOVERNMENT AGENCIES FOR THE PURPOSES OF CONDUCTING THEIR LAWFULLY AUTHORIZED REGULATORY AND ADMINISTRATIVE FUNCTIONS IS SPECIFICALLY ALLOWED.	
2. THE FACILITY IS AN UNMANNED PRIVATE AND SECURED EQUIPMENT INSTALLATION. IT IS ONLY ACCESSED BY TRAINED TECHNICIANS FOR PERIODIC ROUTINE MAINTENANCE AND THEREFORE DOES NOT REQUIRE ANY WATER OR SANITARY SEWER SERVICE. THE FACILITY IS NOT GOVERNED BY REGULATIONS REQUIRING PUBLIC ACCESS PER ADA REQUIREMENTS.	
3. CONTRACTOR SHALL VERIFY ALL PLANS AND EXISTING DIMENSIONS AND CONDITIONS ON THE JOB SITE AND SHALL IMMEDIATELY NOTIFY VIAERO IN WRITING OF DISCREPANCIES BEFORE PROCEEDING WITH THE WORK OR BE RESPONSIBLE FOR SAME.	

PERMITTING PROJECT SUMMARY	
THIS PROJECT CONSISTS OF REMOVAL OF (14) ANTENNAS. REMOVAL OF (14) RRUS. REMOVAL OF (2) MICROWAVE DISHES. INSTALLATION OF (11) ANTENNAS. INSTALLATION OF (12) RRUS. INSTALLATION OF (7) HYBRID CABLES. INSTALLATION OF (2) MICROWAVE DISHES INSTALLATION OF (4) MINILINK 6363 INSTALLATION OF (4) COAX. CABLES. *EXISTING SETBACKS TO REMAIN UNCHANGED*	
PROJECT SUMMARY DETAILS	
REMOVE: ON GROUND (2) DCDU-12B HUAWEI (1) BBU 3910 HUAWEI (1) OPTIX RTN 950 SERIES HUAWEI (1) PANDUIT OPTICOM FIBER TRAY HUAWEI (3) BB6630 (1) OV GMT FUSE PANEL WITH SWING HINGE KIT ON TOWER (8) ADU451819 HUAWEI ANTENNAS (3) AIR 6488 ERICSSON ANTENNAS (3) DBWMXD6512XN HUAWEI ANTENNAS (4) RRU 3971 HUAWEI (8) RRU 3908 V1 HUAWEI (2) RRU 3232 HUAWEI (1) ANDREW MW MICROWAVE ANTENNAS AND RADIO UNITS (1) VHLP4-11W-4WH COMMSCOPE MICROWAVE ANTENNAS AND RADIO UNITS ALL UNUSED FO AND DC CABLES ALL UNUSED COAX CABLES ALL UNUSED TMA'S AND COMBINERS ON TOWER INSTALL: SOW GROUND (1) GPS ANTENNA ERICSSON (2) DCDU BMG907158/1 ERICSSON (LEGACY & COMPACT) (6) RP6647 ERICSSON (3) RBS 6601 ERICSSON (3) BB R503 ERICSSON (2) PSU 48 13 ERICSSON (1) CSR 6675 ERICSSON (1) MINILINK 6692 ERICSSON (4) SPD BOX ERICSSON (1) 12-INCH CABLE LADDER SOW TOWER (8) ROSENBERGER_2D2W45E ANTENNAS (3) AIR 6488 ERICSSON ANTENNAS (3) RRU 4480 B71 LTE 600 MHz (8) RRU 4460 B2 LTE PCS & B66 B4 LTE AWS (4) HYBRID (6x12) CABLES L: 80 METERS (3) HYBRID (3x9) CABLES L: 80 METERS (4) GROUNDING BARS FINAL MW LINK (1) ERICSSON ANTO A 2.4 6L HPX (TR) MW ANTENNA (1) ERICSSON ANT3 A 1.8 6L HPX (DR) MW ANTENNA (4) ERICSSON RADIO/UNIT MINILINK 6363 (4) COAX. CABLES (2x195 FT, 2x165 FT) (2) FLEXIBLE WAVEGUIDES (2) NEW FACE MOUNT PV-TF-M3S3-20	



REVISIONS			
REV.	DATE	DESCRIPTION	BY
A	04/28/23	PRELIMINARY NOT FOR CX	RAS
B	06/17/23	PRELIMINARY NOT FOR CX	RAS
O	06/28/23	FOR CONSTRUCTION	RAS

DRAWING BY: RAS	CHECKED BY: OBN
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SITE NAME: NE-LAWRENCE
SITE NUMBER: 752
SITE ADDRESS: 150 S. SMITH STREET LAWRENCE, NE 88957

SHEET TITLE
TITLE SHEET

SHEET NUMBER
T-1

GENERAL NOTES

1. FOR THE PURPOSE OF CONSTRUCTION DRAWING, THE FOLLOWING DEFINITIONS SHALL APPLY:
CONTRACTOR – GENERAL CONTRACTOR
SUBCONTRACTOR – GENERAL CONTRACTOR (CONSTRUCTION)
OWNER – VIAERO MOBILITY
OEM – ORIGINAL EQUIPMENT MANUFACTURER
2. PRIOR TO THE SUBMISSIONS OF BIDS, THE BIDDING SUBCONTRACTOR SHALL VISIT THE CELL SITE TO FAMILIARIZE THEMSELVES WITH THE EXISTING CONDITIONS AND TO CONFIRM THAT THE WORK CAN BE ACCOMPLISHED AS SHOWN ON THE CONSTRUCTION DRAWINGS. THE CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS, QUANTITIES AND DIMENSIONS BEFORE STARTING ANY WORK. NOTIFY THE CONSTRUCTION MANAGER OF ANY DISCREPANCIES OR INCONSISTENCIES BEFORE PROCEEDING WITH THE WORK.
3. ALL MATERIALS FURNISHED AND INSTALLED SHALL BE IN STRICT ACCORDANCE WITH ALL APPLICABLE CODES, REGULATIONS, AND ORDINANCES. SUBCONTRACTOR SHALL ISSUE ALL APPROPRIATE NOTICES AND COMPLY WITH ALL LAWS, ORDINANCES, RULES, REGULATIONS, AND LAWFUL ORDERS OF ANY PUBLIC AUTHORITY REGARDING THE PERFORMANCE OF THE WORK. ALL WORK CARRIED OUT SHALL COMPLY WITH ALL APPLICABLE MUNICIPAL AND UTILITY COMPANY SPECIFICATIONS AND LOCAL JURISDICTIONAL CODES, ORDINANCES AND APPLICABLE REGULATIONS.
4. DRAWINGS PROVIDED HERE ARE NOT TO BE SCALED AND ARE INTENDED TO SHOW OUTLINE ONLY.
5. UNLESS NOTED OTHERWISE, THE WORK SHALL INCLUDE FURNISHING MATERIALS, EQUIPMENT, APPURTENANCES, AND LABOR NECESSARY TO COMPLETE ALL INSTALLATIONS AS INDICATED ON THE DRAWINGS.
6. THE SUBCONTRACTOR SHALL INSTALL ALL EQUIPMENT AND MATERIALS IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS UNLESS SPECIFICALLY STATED OTHERWISE.
7. IF THE SPECIFIED EQUIPMENT CANNOT BE INSTALLED AS SHOWN ON THESE DRAWINGS, THE SUBCONTRACTOR SHALL PROPOSE AN ALTERNATIVE INSTALLATION FOR APPROVAL BY THE CONTRACTOR.
8. SUBCONTRACTOR SHALL DETERMINE ACTUAL ROUTING OF CONDUIT, POWER AND TI CABLES, GROUNDING CABLES AS SHOWN ON THE POWER, GROUNDING AND TELCO PLAN DRAWING. SUBCONTRACTOR SHALL UTILIZE EXISTING TRAYS AND/OR SHALL ADD NEW TRAYS AS NECESSARY. SUBCONTRACTOR SHALL CONFIRM THE ACTUAL ROUTING WITH THE CONTRACTOR. ROUTING OF TRENCHING SHALL BE APPROVED BY CONTRACTOR.
9. THE SUBCONTRACTOR SHALL PROTECT EXISTING IMPROVEMENTS, PAVEMENTS, CURBS, LANDSCAPING AND STRUCTURES ANY DAMAGED PART SHALL BE REPAIRED AT SUBCONTRACTOR'S EXPENSE TO THE SATISFACTION OF THE OWNER.
10. SUBCONTRACTOR SHALL LEGALLY AND PROPERLY DISPOSE OF ALL SCRAP MATERIALS SUCH AS COAXIAL CABLES AND OTHER ITEMS REMOVED FOR THE EXISTING FACILITY. ANTENNAS REMOVED SHALL BE RETURNED TO THE OWNER'S DESIGNATED LOCATION.
11. SUBCONTRACTOR SHALL LEAVE PREMISES IN CLEAN CONDITION.
12. ALL CONCRETE REPAIR WORK SHALL BE DONE IN ACCORDANCE WITH AMERICAN CONCRETE INSTITUTE (ACI) 301.
13. ANY NEW CONCRETE NEEDED FOR THE CONSTRUCTION SHALL HAVE 4000 PSI STRENGTH AT 28 DAYS UNLESS OTHERWISE SPECIFIED. ALL CONCRETING WORK SHALL BE DONE IN ACCORDANCE WITH ACI 318 CODE REQUIREMENTS.
14. ALL STRUCTURAL STEEL WORK SHALL BE DONE IN ACCORDANCE WITH AISC 13 EDITION SPECIFICATIONS.
15. CONSTRUCTION SHALL COMPLY WITH SPECIFICATION 25741-000-3APS-A00Z-00002, "GENERAL CONSTRUCTION SERVICES".
16. SUBCONTRACTOR SHALL VERIFY ALL EXISTING DIMENSIONS AND CONDITIONS PRIOR TO COMMENCING ANY WORK ALL DIMENSIONS OF EXISTING CONSTRUCTION SHOWN ON THE DRAWINGS MUST BE VERIFIED. SUBCONTRACTOR SHALL NOTIFY THE CONTRACTOR OF ANY DISCREPANCIES PRIOR TO ORDERING MATERIAL OR PROCEEDING WITH CONSTRUCTION.
17. THE EXISTING CELL SITE IS IN FULL COMMERCIAL OPERATION. ANY CONSTRUCTION WORK BY SUBCONTRACTOR SHALL NOT DISRUPT THE EXISTING NORMAL OPERATION. ANY WORK ON EXISTING EQUIPMENT MUST BE COORDINATED WITH CONTRACTOR. ALSO, WORK MAY NEED TO BE SCHEDULED FOR AN APPROPRIATE MAINTENANCE WINDOW USUALLY IN LOW TRAFFIC PERIODS AFTER MIDNIGHT.
18. SINCE THE CELL SITE MAY BE ACTIVE, ALL SAFETY PRECAUTIONS MUST BE TAKEN WHEN WORKING AROUND HIGH LEVELS OF ELECTROMAGNETIC RADIATION. EQUIPMENT SHOULD BE SHUT DOWN PRIOR TO PERFORMING ANY WORK THAT COULD EXPOSE THE WORKERS TO DANGER. PERSONAL RF EXPOSURE MONITORS ARE REQUIRED TO BE WORN TO ALERT OF ANY DANGEROUS EXPOSURE LEVELS.
19. ALL ANTENNA PIPES SHALL BE SCHEDULE 80.
20. LIMITS OF LIABILITY – ITEMS REFERENCED ARE OWNER/CLIENT DICTATED ITEMS, OR SUPPLIED ITEMS WHICH ARE REPRODUCED WITHOUT ALTERATION AS DIRECTED BY OWNER/CLIENT, AND OWNER/CLIENT ASSUMES ANY AND ALL LIABILITY FOR USE OF, CONSEQUENCES OF, OR INTERPRETATION OF SAID ITEM, SPECIFICATION, OR DIRECTIVE; AND AGREES TO INDEMNIFY AND HOLD ENGINEER COMPLETELY HARMLESS.
21. PROFESSIONAL SEAL – DETAILS, SPECIFICATION(S), OR ITEMS REFERENCED, ARE NOT PART OF THE PROFESSIONAL DESIGN PERFORMED BY LICENSEE AND THE PROFESSIONAL SEAL DOES NOT APPLY.

ELECTRICAL INSTALLATION NOTES

1. WIRING, RACEWAY AND SUPPORT METHODS AND MATERIALS SHALL COMPLY WITH THE REQUIREMENTS OF THE NEC AND TELCORDIA.
2. SUBCONTRACTORS SHALL MODIFY EXISTING CABLE TRAY SYSTEM AS REQUIRED TO SUPPORT RF AND TRANSPORT CABLEING TO THE NEW BTS EQUIPMENT. SUBCONTRACTOR SHALL SUBMIT MODIFICATIONS TO CONTRACTOR FOR APPROVAL.
3. ALL CIRCUITS SHALL BE SEGREGATED AND MAINTAIN MINIMUM CABLE SEPARATION AS REQUIRED BY THE NEC AND TELCORDIA.
4. CABLES SHALL NOT BE ROUTED THROUGH LADDER-STYLE CABLE TRAY RUNGS.
5. EACH END OF EVERY POWER, GROUNDING, AND TI CONDUCTOR AND CABLE SHALL BE LABELED WITH COLOR-CODED INSULATION OR ELECTRICAL TAPE (3M BRAND, 1/2 INCH PLASTIC ELECTRICAL TAPE WITH UV PROTECTION, OR EQUAL). THE IDENTIFICATION METHOD SHALL CONFORM WITH NEC & OSHA, AND MATCH EXISTING INSTALLATION REQUIREMENTS.
6. POWER PHASE CONDUCTORS (I.E., HOTS) SHALL BE LABELED WITH COLOR-CODED INSULATION OR ELECTRICAL TAPE (3M BRAND, 1/2 INCH PLASTIC ELECTRICAL TAPE WITH UV PROTECTION, OR EQUAL) PHASE CONDUCTOR COLOR CODES SHALL CONFORM WITH THE NEC & OSHA, AND MATCH EXISTING INSTALLATION REQUIREMENTS.
7. ALL ELECTRICAL COMPONENTS SHALL BE CLEARLY LABELED WITH ENGRAVED LAMACOID PLASTIC LABELS ALL EQUIPMENT SHALL BE LABELED WITH THEIR VOLTAGE RATING, PHASE CONFIGURATION, WIRE CONFIGURATION, POWER OR AMPACITY RATING, AND BRANCH CIRCUIT ID NUMBERS (I.E. , PANELBOARD AND CIRCUIT ID'S).
8. PANELBOARDS (ID NUMBERS) AND INTERNAL CIRCUIT BREAKERS (CIRCUIT ID NUMBERS) SHALL BE CLEARLY LABELED WITH ENGRAVED LAMACOID PLASTIC LABELS.
9. ALL TIE WRAPS WHERE PERMITTED SHALL BE CUT FLUSH WITH APPROVED CUTTING TOOL TO REMOVE SHARP EDGES. USE LOW PROFILES TIE WRAPS.
10. POWER, CONTROL, AND EQUIPMENT GROUND WIRING IN TUBING OR CONDUIT SHALL BE SINGLE CONDUCTOR (12 AWG OR LARGER), 600V, OIL RESISTANT THHN OR THWN-2, CLASS B STRANDED COPPER CABLE RATED FOR 90°C (WET AND DRY) OPERATION; LISTED OR LABELED FOR THE LOCATION AND RACEWAY SYSTEM USED, UNLESS OTHERWISE SPECIFIED.
11. SUPPLEMENTAL EQUIPMENT GROUND WIRING LOCATED INDOORS SHALL BE SINGLE CONDUCTOR (6 AWG OR LARGER), 600V, OIL RESISTANT THHN OR THWN-2, CLASS B STRANDED COPPER CABLE RATED FOR 90°C (WET AND DRY) OPERATION; LISTED OR LABELED FOR THE LOCATION AND RACEWAY SYSTEM USED, UNLESS OTHERWISE SPECIFIED.
12. SUPPLEMENTAL EQUIPMENT GROUND WIRING LOCATED OUTDOORS, OR BELOW GRADE, SHALL BE SINGLE CONDUCTOR 2 AWG SOLID TINNED COPPED CABLE, UNLESS OTHERWISE SPECIFIED.
13. POWER WIRING, NOT IN TUBING OR CONDUIT, SHALL BE MULTI-CONDUCTOR, TYPE TC CABLE (12 AWG OR LARGER), 600V, OIL RESISTANT THHN OR THWN-2, CLASS B STRANDED COPPER CABLE RATED FOR 90°C (WET AND DRY) OPERATION; LISTED OR LABELED FOR THE LOCATION USED, UNLESS OTHERWISE SPECIFIED.
14. ALL POWER AND GROUNDING CONNECTIONS SHALL BE CRIMP-STYLE, COMPRESSION WIRE LUGS AND WIRENUTS BY THOMAS AND BETTS (OR EQUAL) LUGS AND WIRENUTS SHALL BE RATED FOR OPERATION AT ON LESS THAN 75°C (90°C IF AVAILABLE).
15. RACEWAY AND CABLE TRAY SHALL BE LISTED OR LABELED FOR ELECTRICAL USE IN ACCORDANCE WITH NEMA, UL, ANSI/IEEE, AND NEC.
16. NEW RACEWAY OR CABLE TRAY WILL MATCH THE EXISTING INSTALLATION WHERE POSSIBLE.
17. ELECTRICAL METALLIC TUBING (EMT) OR RIGID NONMETALLIC CONDUIT (I.E., RIGID PVC SCHEDULE 40, OR RIGID PVC SCHEDULE 80 FOR LOCATIONS SUBJECT TO PHYSICAL DAMAGE) SHALL BE USED FOR EXPOSED INDOOR LOCATIONS.
18. ELECTRICAL METALLIC TUBING (EMT) OR ELECTRICAL NONMETALLIC TUBING (ENT), OR RIGID NONMETALLIC CONDUIT (RIGID PVC SCHEDULE 40) SHALL BE USED FOR CONCEALED INDOOR LOCATIONS.
19. GALVANIZED STEEL INTERMEDIATE METALLIC CONDUIT (IMC) SHALL BE USED FOR OUTDOOR LOCATIONS ABOVE GRADE.
20. RIGID NONMETALLIC CONDUIT (I.E., RIGID PVC SCHEDULE 40, OR RIGID PVC SCHEDULE 80) SHALL BE USED UNDERGROUND; DIRECT BURIED, IN AREAS OF OCCASIONAL LIGHT VEHICLE TRAFFIC OR ENCASED IN REINFORCED CONCRETE IN AREAS OF HEAVY VEHICLE TRAFFIC.
21. LIQUID-TIGHT FLEXIBLE METALLIC CONDUIT (LIQUID-TITE FLEX) SHALL BE USED INDOORS AND OUTDOORS, WHERE VIBRATION OCCURS OR FLEXIBILITY IS NEEDED.
22. CONDUIT AND TUBING FITTINGS SHALL BE THREADED OR COMPRESSION-TYPE AND APPROVED FOR THE LOCATION USED. SETSCREW FITTINGS ARE NOT ACCEPTABLE.
23. CABINETS, BOXES, AND WIREWAYS SHALL BE LISTED OR LABELED FOR ELECTRICAL USE IN ACCORDANCE WITH NEMA, UL, ANSI/IEEE AND NEC.
24. CABINETS, BOXES AND WIREWAYS TO MATCH THE EXISTING INSTALLATION WHERE POSSIBLE.
25. WIREWAYS SHALL BE EPOXY-COATED (GRAY) AND INCLUDE A HINGED COVER, DESIGNED TO SWING OPEN DOWNWARD; SHALL BE PANDUIT TYPE E (OR EQUAL); AND RATED NEMA 1 (OR BETTER) INDOORS, OR NEMA 34 (OR BETTER) OUTDOORS.
26. EQUIPMENT CABINETS, TERMINAL BOXES, JUNCTION BOXES, AND PULL BOXES SHALL BE GALVANIZED OR EPOXY-COATED SHEET STEEL, SHALL MEET OR EXCEED UL 50, AND RATED NEMA 1 (OR BETTER) INDOORS, OR NEMA 34 (OR BETTER) OUTDOORS.
27. METAL RECEPTACLE, SWITCH, AND DEVICE BOXES SHALL BE GALVANIZED, EPOXY-COATED, OR NON-CORRODING; SHALL MEET OR EXCEED UL 514A AND NEMA OS 1; AND RATED NEMA 1 (OR BETTER) INDOORS, OR WEATHER PROTECTED (WP OR BETTER) OUTDOORS.
28. NONMETALLIC RECEPTACLE, SWITCH, AND DEVICE BOXES SHALL MEET OR EXCEED NEMA OS 2; AND RATED NEMA 1 (OR BETTER) INDOORS, OR WEATHER PROTECTED (WP OR BETTER) OUTDOORS.
29. THE SUBCONTRACTOR SHALL NOTIFY AND OBTAIN NECESSARY AUTHORIZATION FROM THE CONTRACTOR BEFORE COMMENCING WORK ON THE AC POWER DISTRIBUTION PANELS.
30. THE SUBCONTRACTOR SHALL PROVIDE NECESSARY TAGGING ON THE BREAKERS, CABLES AND DISTRIBUTION PANELS IN ACCORDANCE WITH THE APPLICABLE CODES AND STANDARDS TO SAFEGUARD AGAINST LIFE AND PROPERTY.

GROUNDING NOTES

1. THE SUBCONTRACTOR SHALL REVIEW AND INSPECT THE EXISTING FACILITY GROUNDING SYSTEM AND LIGHTNING PROTECTION SYSTEM (AS DESIGNED AND INSTALLED) FOR STRICT COMPLIANCE WITH THE NEC (AS ADOPTED BY THE AHJ). THE SITE-SPECIFIC (UL, LPI, OR NFPA) LIGHTING PROTECTION CODE, AND GENERAL COMPLIANCE WITH TELCORDIA AND TIA GROUNDING STANDARDS. THE SUBCONTRACTOR SHALL REPORT ANY VIOLATIONS OR ADVERSE FINDINGS TO THE CONTRACTOR FOR RESOLUTION.
2. ALL GROUND ELECTRODE SYSTEMS (INCLUDING TELECOMMUNICATION, RADIO, LIGHTNING PROTECTION, AND AC POWER GES'S) SHALL BE BONDED TOGETHER, AT OR BELOW GRADE, BY TWO OR MORE COPPER BONDING CONDUCTORS IN ACCORDANCE WITH THE NEC.
3. THE SUBCONTRACTOR SHALL PERFORM IEEE FALL-OF-POTENTIAL RESISTANCE TO EARTH TESTING (PER IEEE 1100 AND 91) FOR NEW GROUND ELECTRODE SYSTEMS. THE SUBCONTRACTOR SHALL FURNISH AND INSTALL SUPPLEMENTAL GROUND ELECTRODES AS NEEDED TO ACHIEVE A TEST RESULT OF 5 OHMS OR LESS. TESTS SHALL BE PERFORMED IN ACCORDANCE WITH 25471-000-3PS-EG00-0001, DESIGN & TESTING OF FACILITY GROUNDING FOR CELL SITES.
4. METAL RACEWAY SHALL NOT BE USED AS THE NEC REQUIRED EQUIPMENT GROUND CONDUCTOR. STRANDED COPPER CONDUCTORS WITH GREEN INSULATION, SIZED IN ACCORDANCE WITH THE NEC, SHALL BE FURNISHED AND INSTALLED WITH THE POWER CIRCUITS TO BTS EQUIPMENT.
5. EACH BTS CABINET FRAME SHALL BE DIRECTLY CONNECTED TO THE MASTER GROUND BAR WITH GREEN INSULATED SUPPLEMENTAL EQUIPMENT GROUND WIRES, 6 AWG STRANDED COPPER OR LARGER INDOORS BTS; 2 AWG STRANDED COPPER FOR OUTDOORS BTS.
6. EXOTHERMIC WELDS SHALL BE USED FOR ALL GROUNDING CONNECTIONS BELOW GRADE.
7. APPROVED ANTIOXIDANT COATINGS (I.E., CONDUCTIVE GEL OR PASTE) SHALL BE USED ON ALL COMPRESSION AND BOLTED GROUND CONNECTIONS.
8. ICE BRIDGE BONDING CONDUCTORS SHALL BE EXOTHERMICALLY BONDED OR BOLTED WITH STAINLESS STEEL HARDWARE TO THE BRIDGE AND THE TOWER GROUND BAR.
9. ALUMINUM CONDUCTOR OR COPPER CLAD STEEL CONDUCTOR SHALL NOT BE USED FOR GROUNDING CONNECTIONS.
10. MISCELLANEOUS ELECTRICAL AND NON-ELECTRICAL METAL BOXES, FRAMES AND SUPPORTS SHALL BE BONDED TO THE GROUND RING, IN ACCORDANCE WITH THE NEC.
11. METAL CONDUIT AND TRAY SHALL BE GROUNDED AND MADE ELECTRICALLY CONTINUOUS WITH LISTED BONDING FITTINGS OR BY BONDING ACROSS THE DISCONTINUITY WITH 6 AWG COPPER WIRE UL APPROVED GROUNDING TYPE CONDUIT CLAMPS.
12. GROUND CONDUCTORS USED IN THE FACILITY GROUND AND LIGHTNING PROTECTION SYSTEMS SHALL NOT BE ROUTED THROUGH METALLIC OBJECTS THAT FORM A RING AROUND THE CONDUCTOR, SUCH AS METALLIC CONDUITS, METAL SUPPORT CLIPS OR SLEEVES THROUGH WALLS OR FLOORS. WHEN IT IS REQUIRED TO BE HOUSED IN CONDUIT TO MEET CODE REQUIREMENTS OR LOCAL CONDITIONS, NON-METALLIC MATERIAL, SUCH AS PVC PLASTIC CONDUIT SHALL BE USED. WHERE USE OF METAL CONDUIT IS UNAVOIDABLE (E.G., NON-METALLIC CONDUIT PROHIBITED BY LOCAL CODE) THE GROUND CONDUCTOR SHALL BE BONDED TO EACH END OF THE METAL CONDUIT.
13. ALL TOWER GROUND SYSTEMS SHALL COMPLY WITH THE REQUIREMENTS OF ANSI/TIA 222 FOR TOWERS BEING BUILT TO REV G OF THE STANDARD, THE WIRE SIZE OF THE BURIED GROUND RING AND CONNECTIONS BETWEEN THE TOWER AND THE BURIED GROUND RING SHALL BE CHANGED FROM 2 AWG TO 2/0 AWG. IN ADDITION, THE MINIMUM LENGTH OF THE GROUND RODS SHALL BE INCREASED FOR 8 FEET TO 10 FEET.
14. ALL GROUND WIRE TO RRUS SHALL BE #2 GREEN STRANDED.
15. ALL OUTDOOR LUGS SHALL USE BLACK HEAT SHRINK AND INDOOR LUGS SHALL USE CLEAR HEAT SHRINK.
16. ALL OUTDOOR LUGS TO BE LONG BARREL 2 HOLE WITHOUT INSPECTION HOLES AND INDOOR LUGS TO HAVE INSPECTION HOLES.

ABBREVIATIONS

AGL	ABOVE GRADE LEVEL	MFR	MANUFACTURER
AMSL	ABOVE MEAN SEA LEVEL	MGB	MASTER GROUND BAR
AWG	AMERICAN WIRE GAUGE	MIN	MINIMUM
BLDG	BUILDING	MW	MICROWAVE
DWG	DRAWING	N.T.S.	NOT TO SCALE
FT	FOOT	(P)	PROPOSED
EMT	ELECTRICAL METALLIC TUBING	PPC	POWER PROTECTION CABINET
ELEV	ELEVATION	RBS	RADIO BASE STATION
EQUIP	EQUIPMENT	RRU	REMOTE RADIO UNIT
(E)	EXISTING	SPD	SURGE PROTECTION DEVCIE
EXT	EXTERIOR	TDD	TIME DIVISION DUPLEX
FND	FOUNDATION	IN	INCH(ES)
FDD	FREQUENCY DIVISION DUPLEX	INT	INTERIOR
FO	FIBER OPTIC	LB(S) OR #	POUND(S)
GALV	GALVANIZED	SF	SQUARE FOOT
GPS	GLOBAL POSITIONING SYSTEM	TYP	TYPICAL
GND	GROUND	W/	WITH
LTE	LONG TERM EVOLUTION	XFMR	TRANSFORMER
MAX	MAXIMUM		



VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

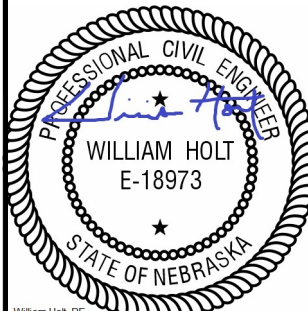


6300 LEGACY DRIVE
PLANO, TX 75024



319 CHAPMAN RD, SUITE 118
RALEIGH, NC 27603
PH: (405)348-5460 FAX: (405)341-4825

06/29/2023



William Holt, PE
Director of Engineering
License No. E-18973 Expires: 12/31/2023
COA No. CA2573 Expires: 08/08/2023

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B	06/17/23	PRELIMINARY NOT FOR CX	RAS
O	06/28/23	FOR CONSTRUCTION	RAS

DRAWING BY: RAS CHECKED BY: OBN

SITE NAME:
NE-LAWRENCE

SITE NUMBER:
752

SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68597

SHEET TITLE

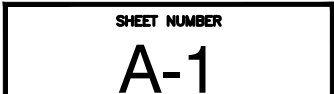
GENERAL NOTES

SHEET NUMBER

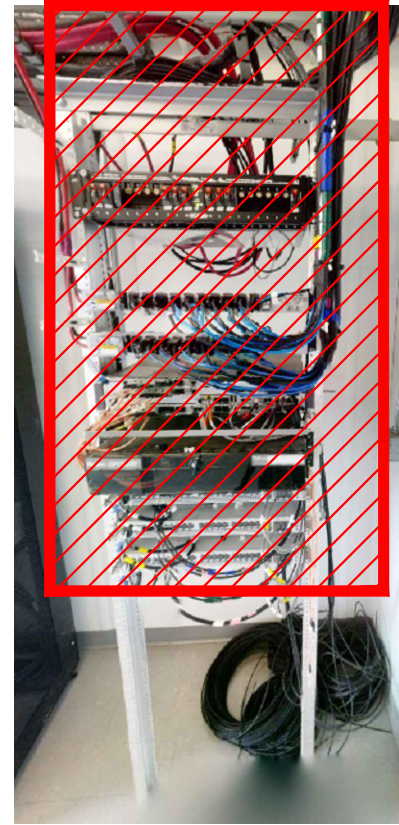
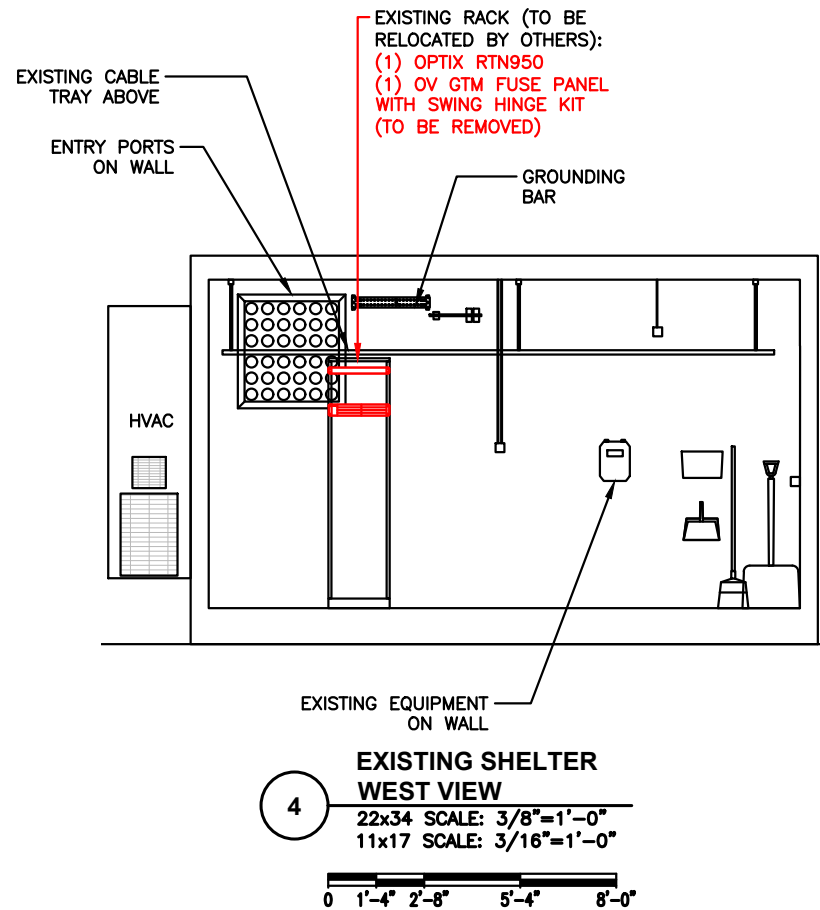
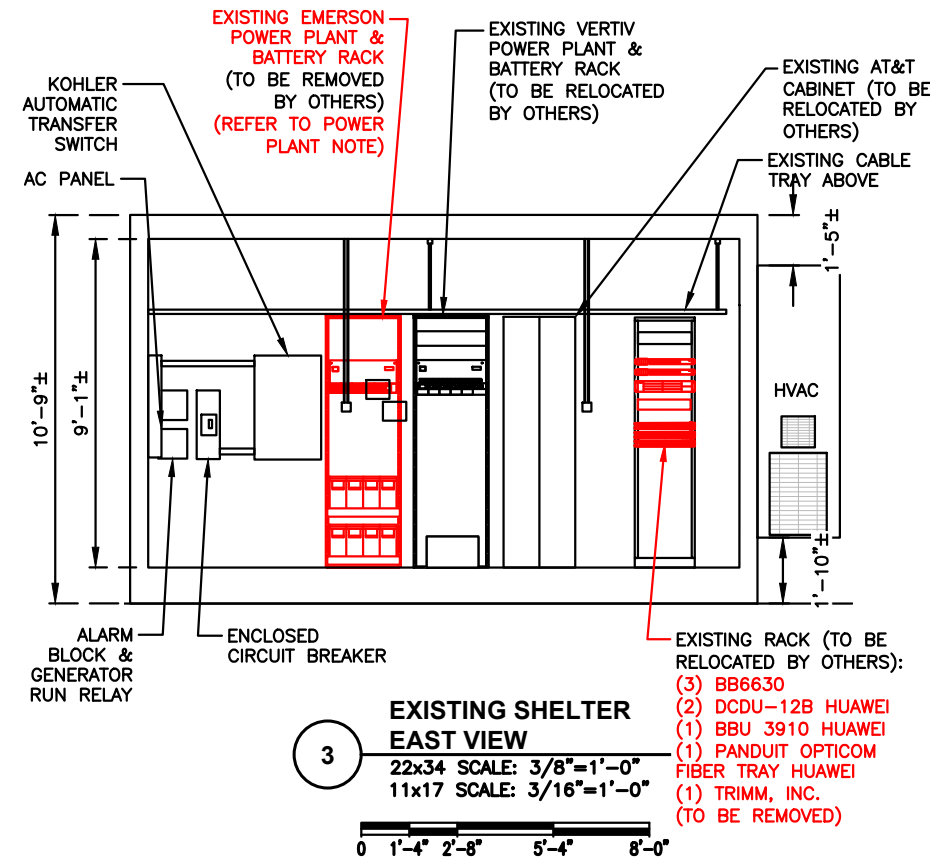
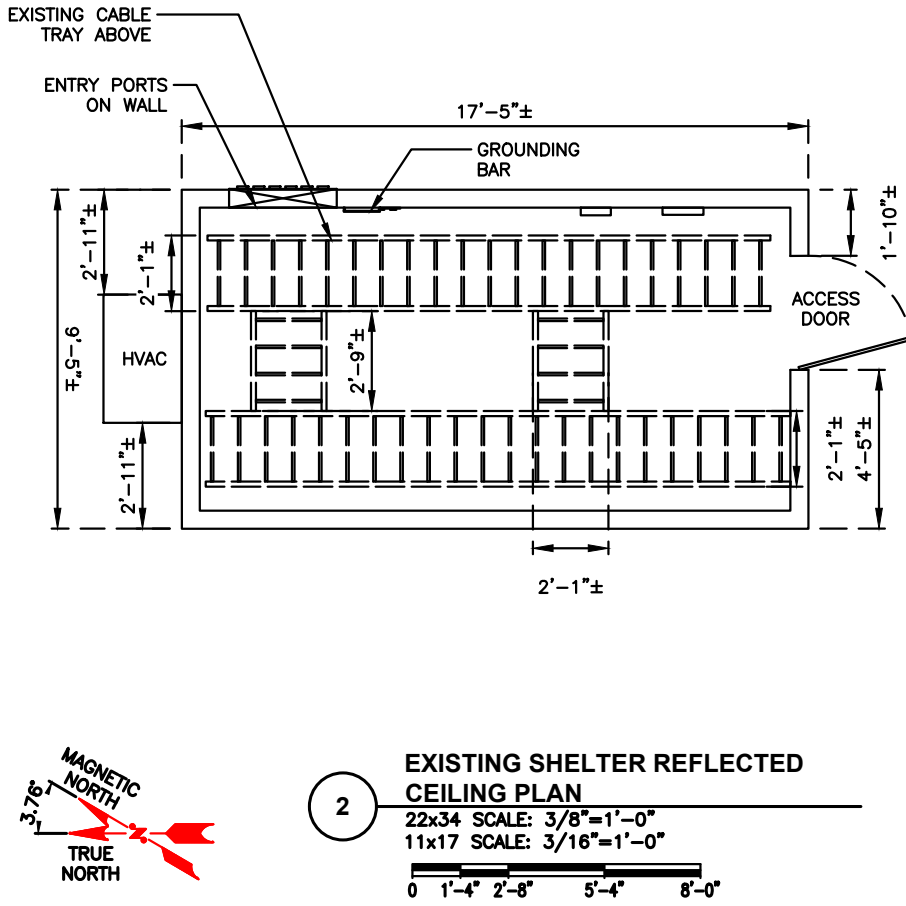
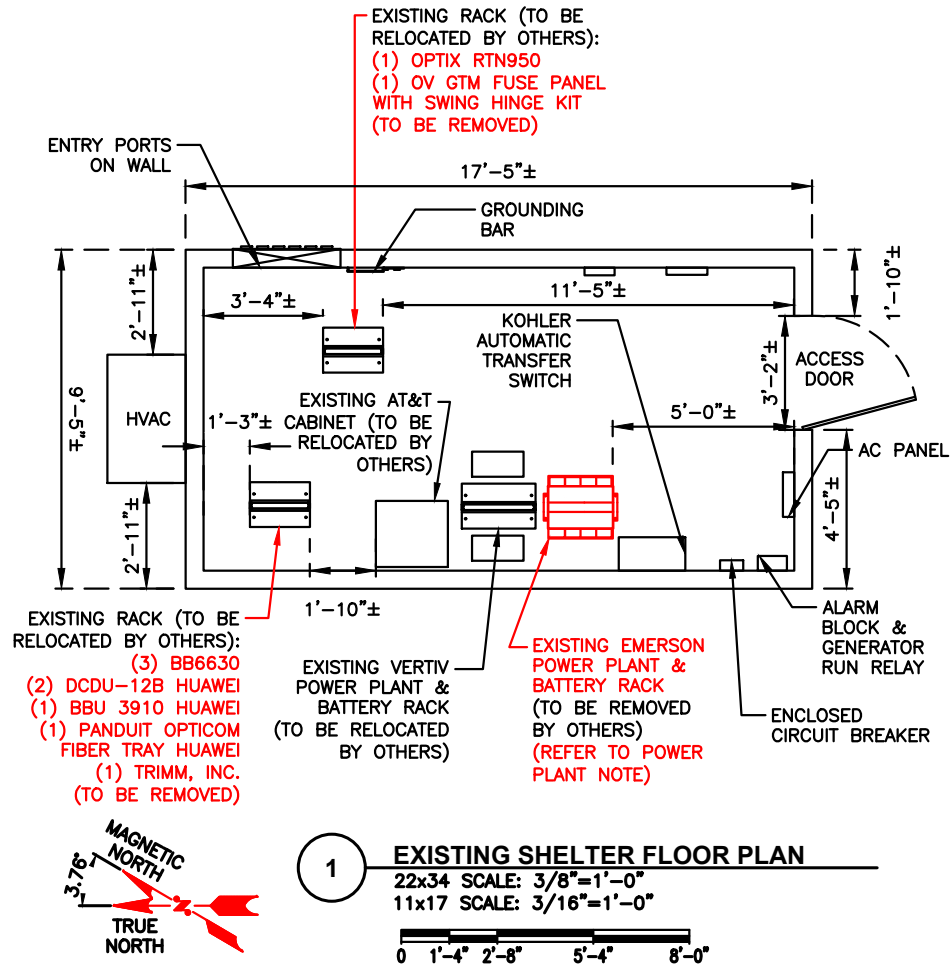
GN-1



NOTE:
REFER TO A-15 & A-16
DATA SHEET FOR FINAL
EQUIPMENT INFORMATION.

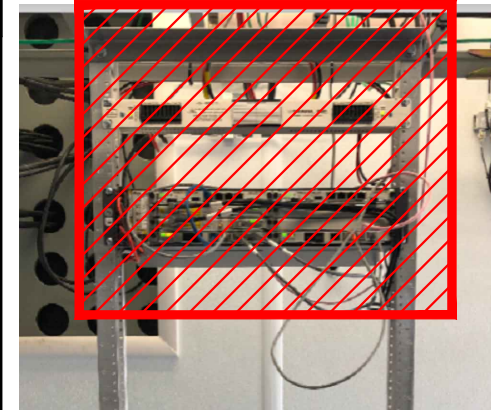


752_NE-LAWRENCE_REVO_062823.DWG

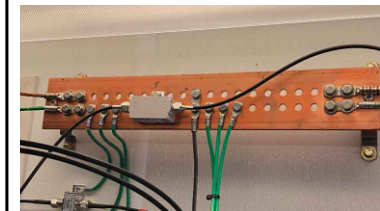


POWER PLANT NOTE:
EXISTING POWER PLANT & BATTERY RACK WILL BE REMOVED BY OTHERS AS PART OF ANOTHER PROJECT PRIOR INSTALLING PWR 6230.

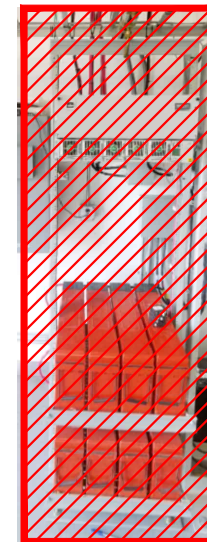
REMOVE: (2)
DCDU-12B, (1)
BBU3910, (1)
PANDUIT OPTICOM
FIBER TRAY HUAWEI,
(1) TRIMM, INC.
(3) BB6630



REMOVE: (1)
OPTIX RTN 950
SERIES, (1) OV
GTM FUSE
PANEL WITH
SWING HINGE KIT



EXISTING GROUNDING BAR



REMOVE: (1) EMERSON
POWER PLANT &
BATTERY RACK

VIAERO
WIRELESS

VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

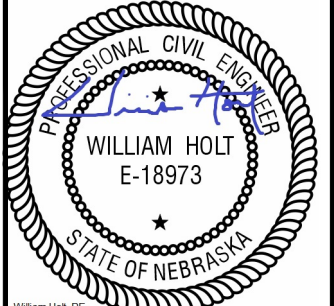
ERICSSON

6300 LEGACY DRIVE
PLANO, TX 75024

telamon.CLS

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NE-LAWRENCE

SITE NUMBER:
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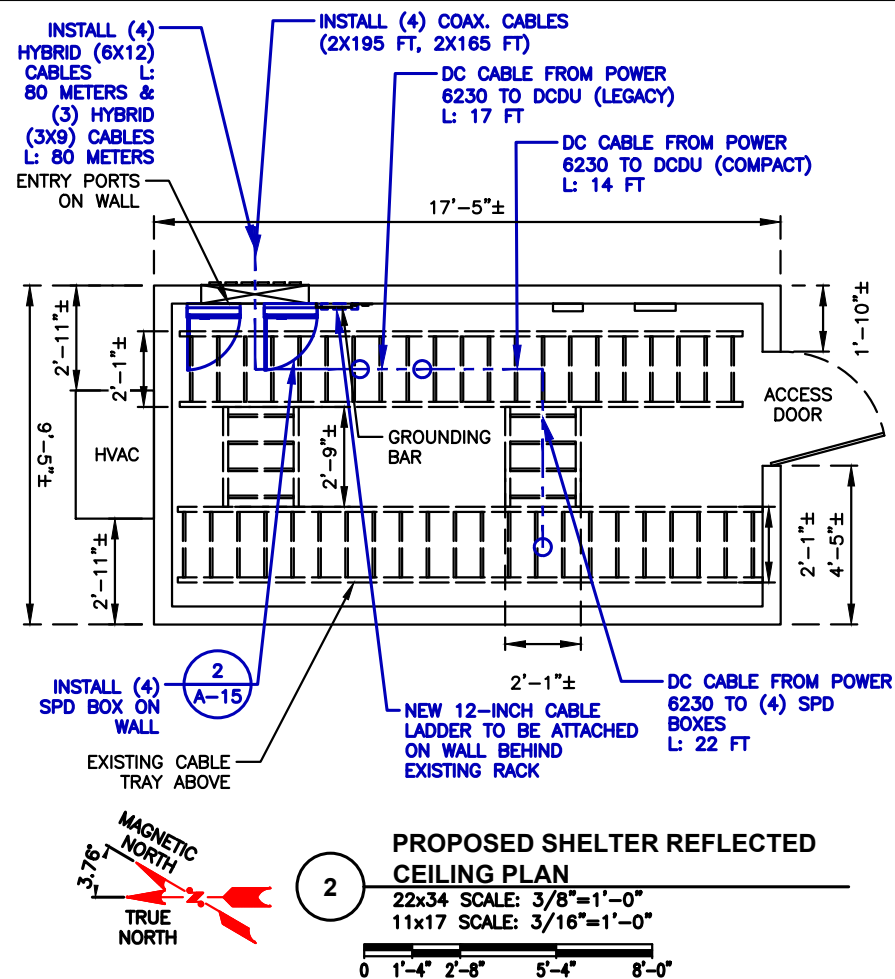
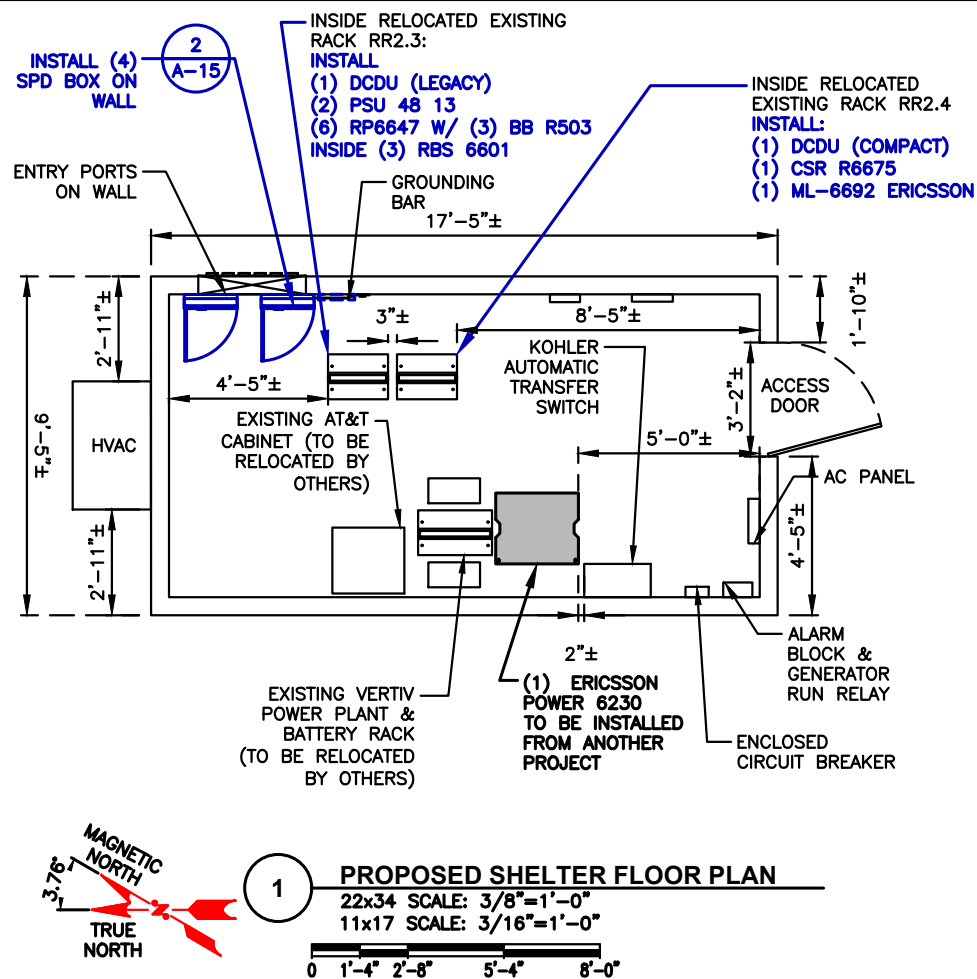
SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE

EXISTING EQUIPMENT
SHELTER DETAILS

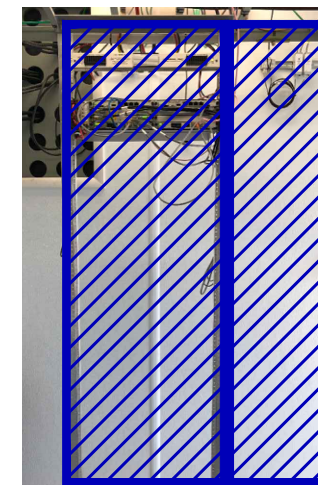
SHEET NUMBER

A-2



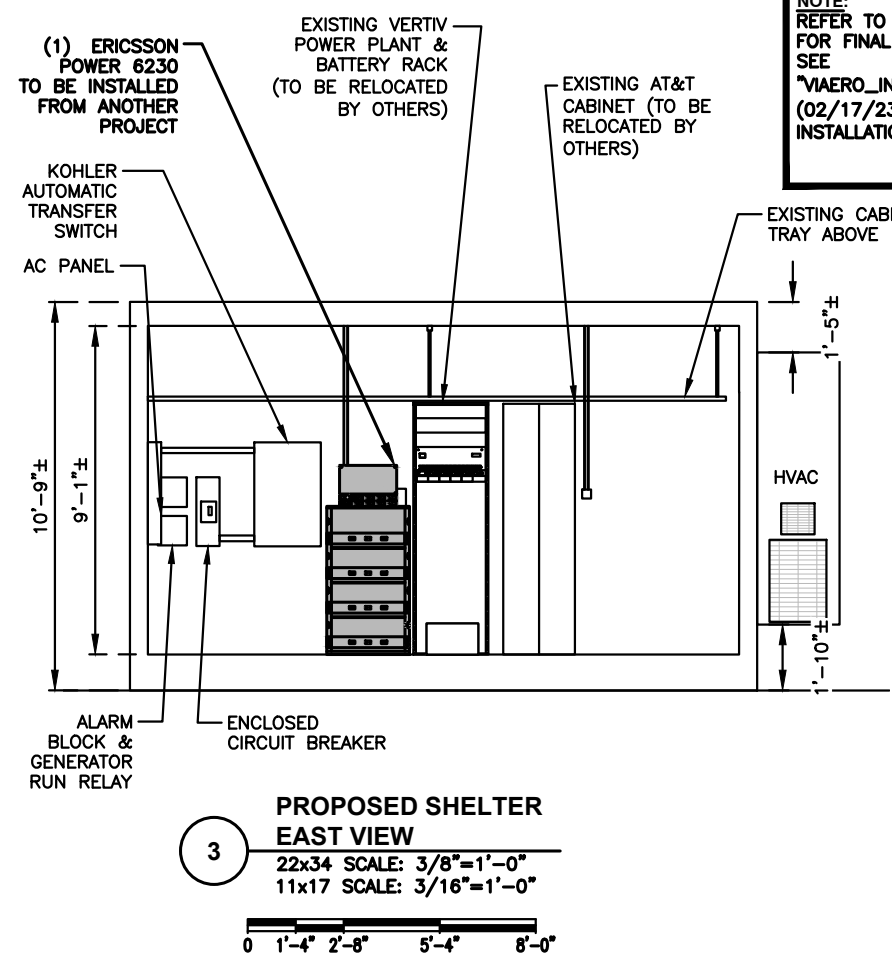
SPECIFIC MOUNTING POSITION IN RACK RR2.4 FOR MW EQUIPMENT:
 RRU45=DCDU (COMPACT)
 RRU43=CSR R6675
 RRU42=FIBER SHELF FOR CSR
 RRU41/40/39=ML-6692 TO NE-BLUE_HILL
 RRU37=FIBER SHELF FOR ML

SPECIFIC MOUNTING POSITION IN RACK RR2.3 FOR RF EQUIPMENT:
 RRU45=PSU 4813
 RRU44=PSU 4813
 RRU43=DCDU (LEGACY)
 RRU41/40=PANDUIT FRUE2U
 RRU39=RP6647 #1
 RRU37/36=RP6647 #1 W/ BBR503 INSIDE RBS 6601 #1
 RRU34=RP6647 #2
 RRU32/31=RP6647 #2 W/ BBR503 INSIDE RBS 6601 #2
 RRU29=RP6647 #3
 RRU27/26=RP6647 #3 W/ BBR503 INSIDE RBS 6601 #3
 RRU24=BB RP6647 #4
 RRU22=BB RP6647 #5
 RRU20=BB RP6647 #6

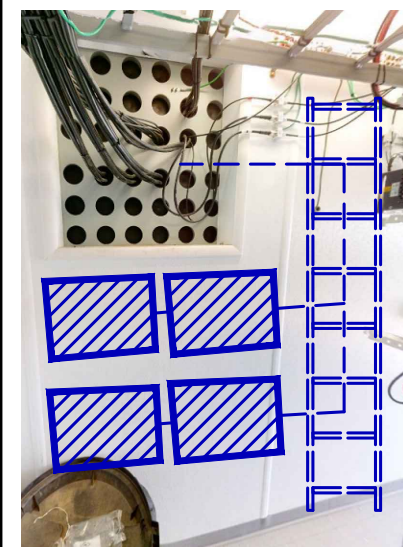
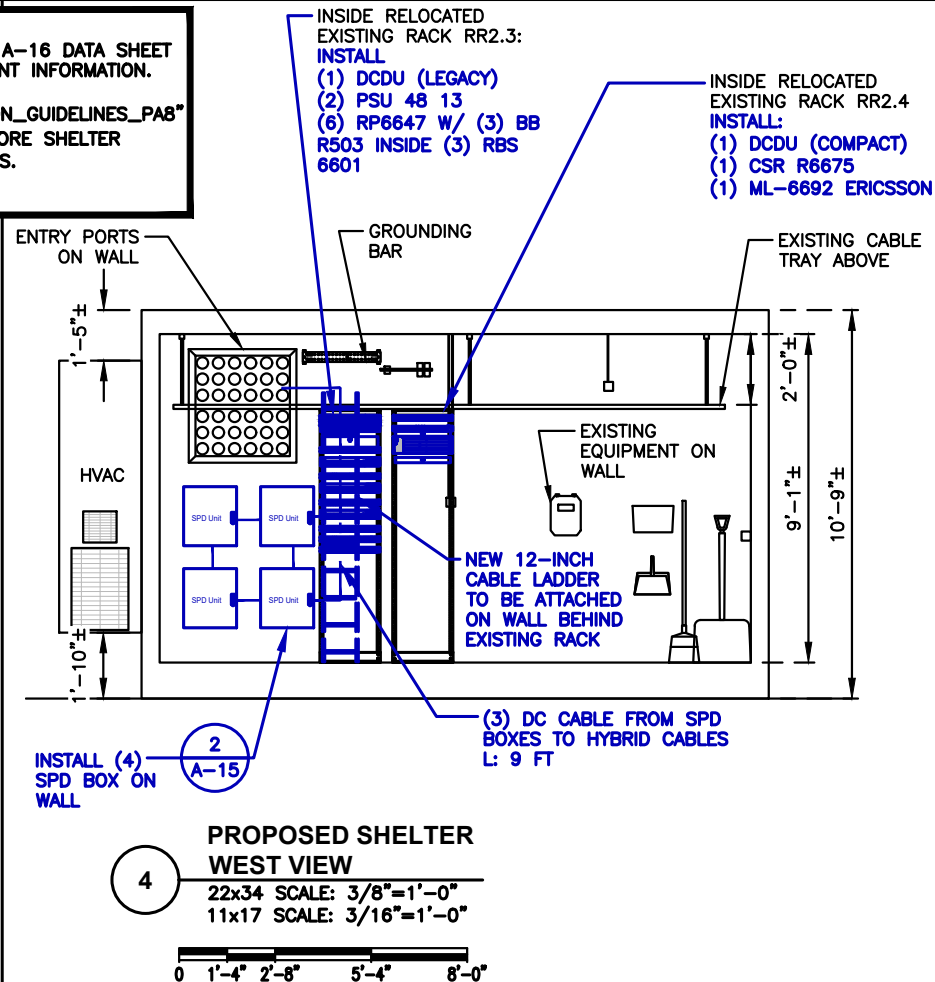


INSTALL: (2) PSU 4813, (1) DCDU (LEGACY), (6) BB RP6647 W/ (3) BB R503 INSIDE (3) RBS 6601

INSTALL: (1) DCDU (COMPACT), (1) CSR R6675 & (1) ML-6692 ERICSSON



NOTE:
 REFER TO A-15 & A-16 DATA SHEET FOR FINAL EQUIPMENT INFORMATION. SEE "VIAERO_INSTALLATION_GUIDELINES_PA8" (02/17/23) FOR MORE SHELTER INSTALLATION DETAILS.



INSTALL: (4) SPD BOXES MOUNTED ON WALL & CABLE LADDER ATTACHED TO THE WALL



TO BE INSTALLED FROM ANOTHER PROJECT: (1) PW6230

VIAERO WIRELESS

VIAERO WIRELESS
 1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

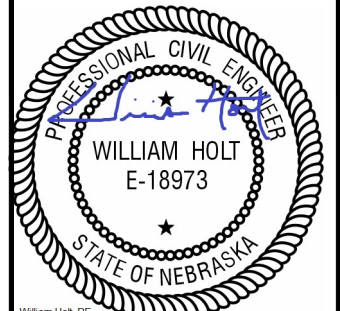
ERICSSON

6300 LEGACY DRIVE
 PLANO, TX 75024

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DRAWING BY: RAS CHECKED BY: OBN

SITE NAME:
 NE-LAWRENCE

SITE NUMBER:
 752

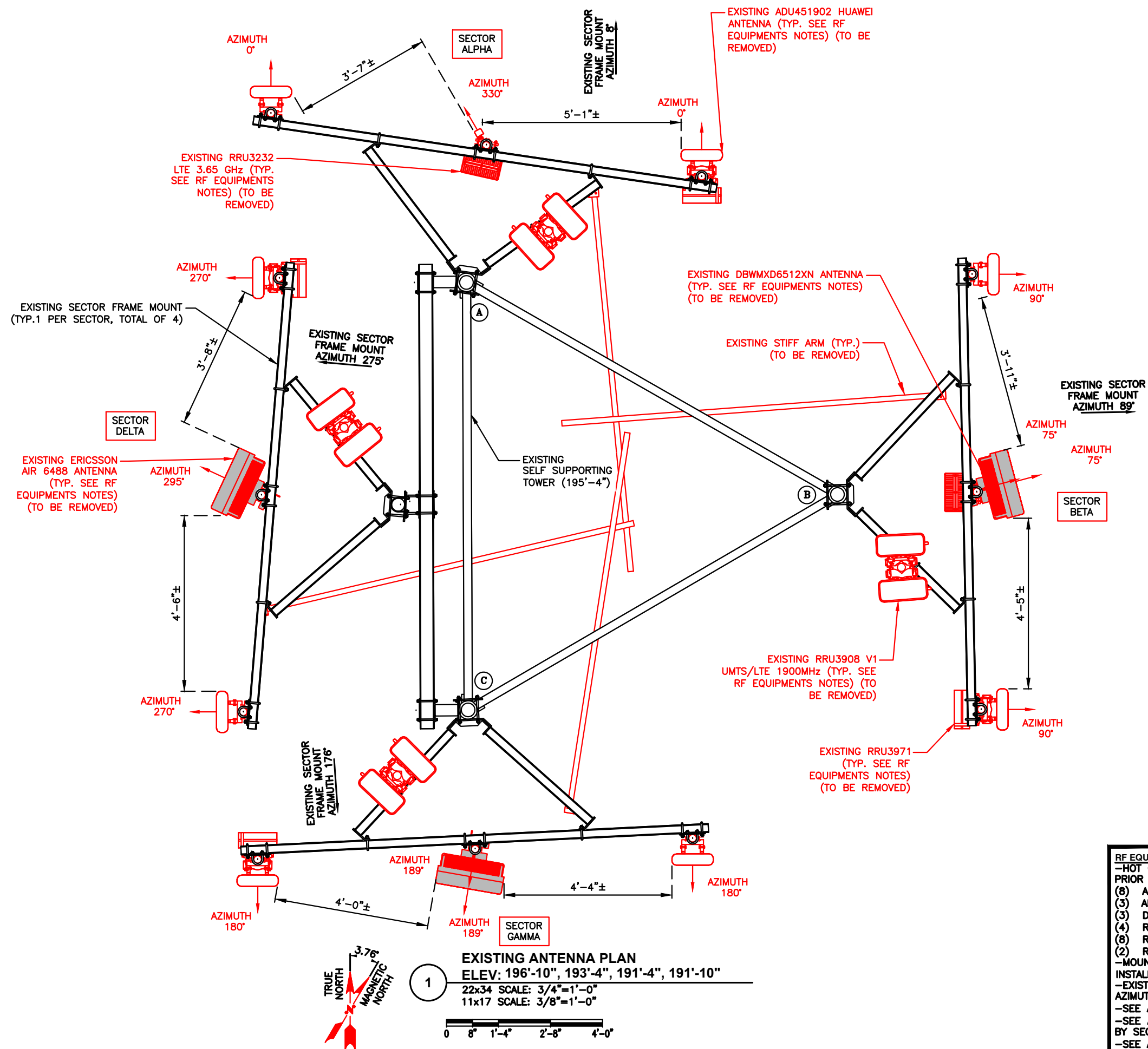
SITE ADDRESS:
 150 S. SMITH STREET
 LAWRENCE, NE 68957

SHEET TITLE

PROPOSED
 EQUIPMENT SHELTER
 DETAILS

SHEET NUMBER

A-3



RF EQUIPMENT NOTE:
 -HOT CUT SITE. EXISTING EQUIPMENT ON TOWER WILL BE REMOVED PRIOR TO THE INSTALLATION OF THE NEW ERICSSON EQUIPMENT.
 (8) ADU451819 HUAWEI ANTENNAS AT 193'-4"
 (3) AIR 6488 ERICSSON ANTENNAS AT 191'-4"(1) & 191'-10"(2)
 (3) DBWMXD6512XN HUAWEI ANTENNAS AT 196'-10"
 (4) RRU 3971 HUAWEI AT 194'-8"
 (8) RRU 3908 V1 HUAWEI AT 195'-4"
 (2) RRU 3232 HUAWEI AT 194'-8"
 -MOUNT MODS MUST BE INSTALLED PRIOR TO NEW EQUIPMENT BEING INSTALLED (SEE A-6)
 -EXISTING MOUNT FRAMES TO BE TRUED UP TO NEW RF EQUIPMENT AZIMUTH.
 -SEE A-7 & A-11 FOR FINAL PROPOSAL, PLAN AND ELEVATION.
 -SEE A-12 FOR NUMBER OF ANTENNAS & RADIOS TO BE INSTALLED BY SECTOR AND ELEVATION.
 -SEE A-15 FOR NEW EQUIPMENT DIMENSIONS AND DATA.



VIAERO WIRELESS
1224 WEST PLATTE AVENUE FORT MORGAN, CO 80701

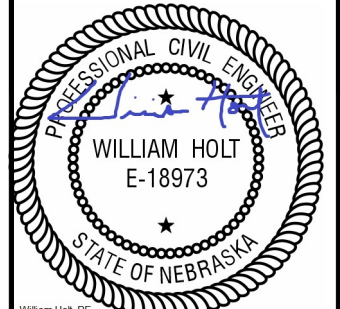


6300 LEGACY DRIVE
PLANO, TX 75024



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REVISIONS

[illegible]

DRAWING BY: RAS	CHECKED BY: OBN
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SITE NAME:
NE-LAWRENCE

SITE NUMBER:
752

SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 88957

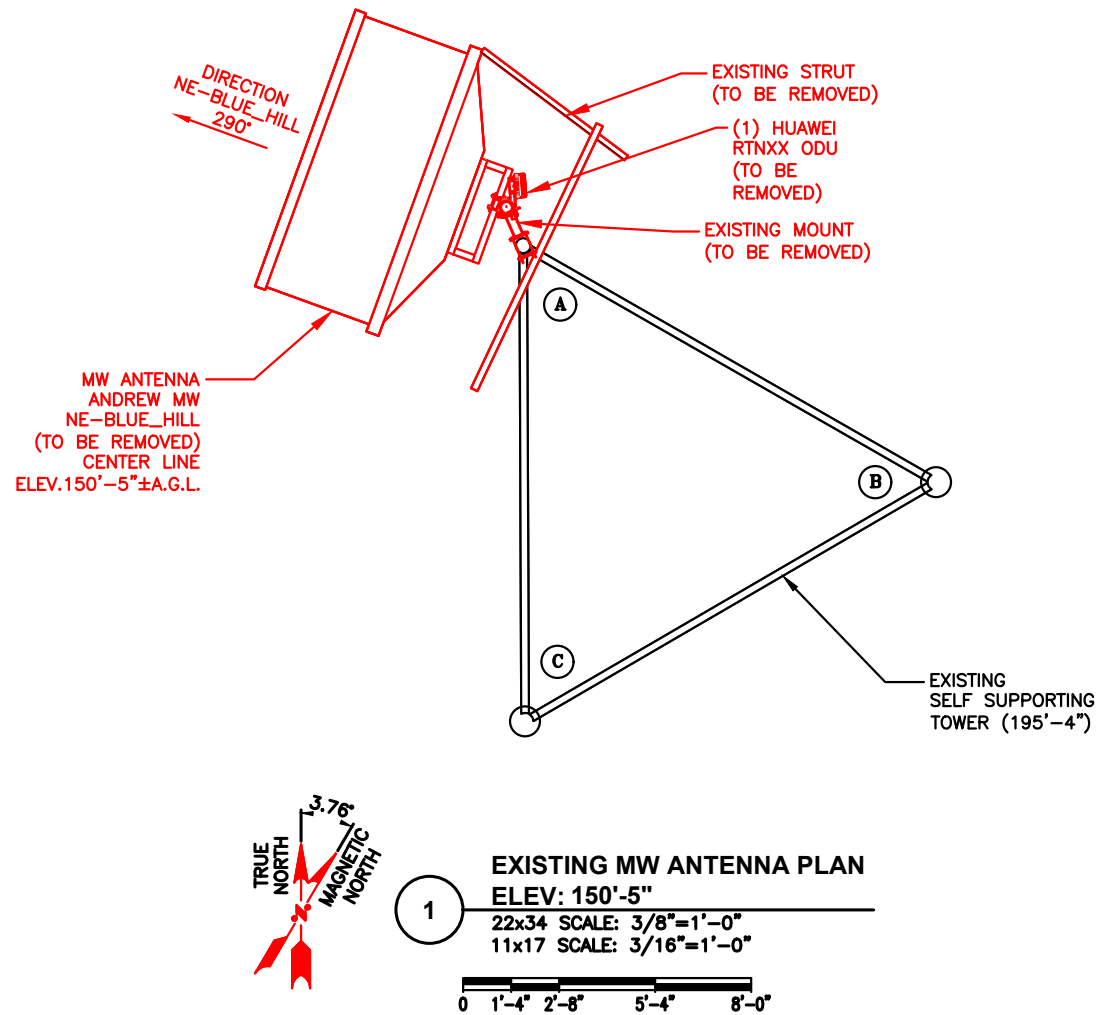
SHEET TITLE

EXISTING ANTENNA
PLAN VIEW ELEV:
196'-10", 193'-4",
191'-4", 191'-10"

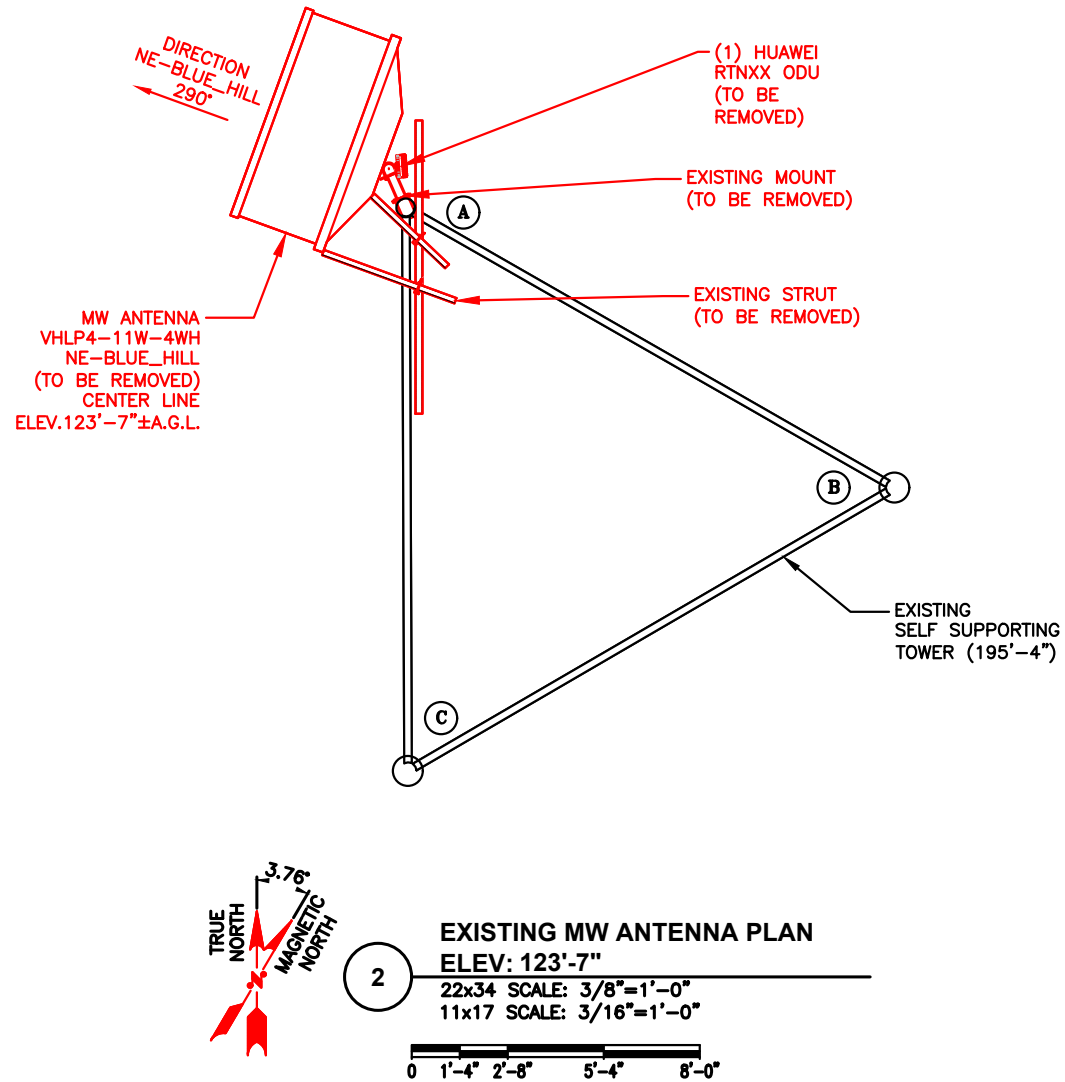
SHEET NUMBER

A-4

752_NE-LAWRENCE_REVO_062823.DWG



MW ANTENNA AT
150'-5"



MW ANTENNA AT
123'-7"

VIAERO
WIRELESS

VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

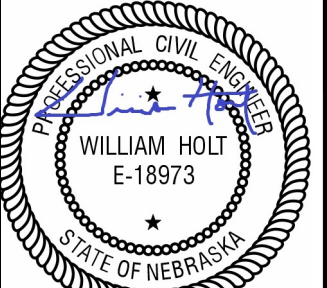
ERICSSON

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PLANO, TX 75024

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DRAWING BY: RAS CHECKED BY: OBN

SITE NAME:
NE-LAWRENCE

SITE NUMBER:
752

SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE

EXISTING MW ANTENNA
PLAN VIEW ELEV:
150'-5", 123'-7"

SHEET NUMBER

A-5

752_NE-LAWRENCE_REVO_062823.DWG

HOT CUT NOTE:
-THERE IS NOT ENOUGH SPACE TO
INSTALL THE EXISTING AND NEW
EQUIPMENT ON THE SAME MOUNT.

RF EQUIPMENT NOTE:
-HOT CUT SITE. EXISTING EQUIPMENT ON TOWER WILL BE REMOVED
PRIOR TO THE INSTALLATION OF THE NEW ERICSSON EQUIPMENT.
(8) ADU451819 HUAWEI ANTENNAS AT 193'-4"
(3) AIR 6488 ERICSSON ANTENNAS AT 191'-4"(1) & 191'-10"(2)
(3) DBWMXD6512XN HUAWEI ANTENNAS AT 196'-10"
(4) RRU 3971 HUAWEI AT 194'-8"
(8) RRU 3908 V1 HUAWEI AT 195'-4"
(2) RRU 3232 HUAWEI AT 194'-8"
-MOUNT MODS MUST BE INSTALLED PRIOR TO NEW EQUIPMENT
BEING INSTALLED (SEE A-6)
-EXISTING MOUNT FRAMES TO BE TRUED UP TO NEW RF
EQUIPMENT AZIMUTH.
-SEE A-7 & A-11 FOR FINAL PROPOSAL, PLAN AND ELEVATION.
-SEE A-12 FOR NUMBER OF ANTENNAS & RADIOS TO BE
INSTALLED BY SECTOR AND ELEVATION.
-SEE A-15 FOR NEW EQUIPMENT DIMENSIONS AND DATA.



VIAERO WIRELESS
1224 WEST PLATTE AVENUE FORT MORGAN, CO 80701

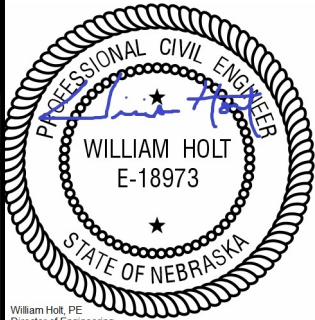


6300 LEGACY DRIVE
PLANO, TX 75024



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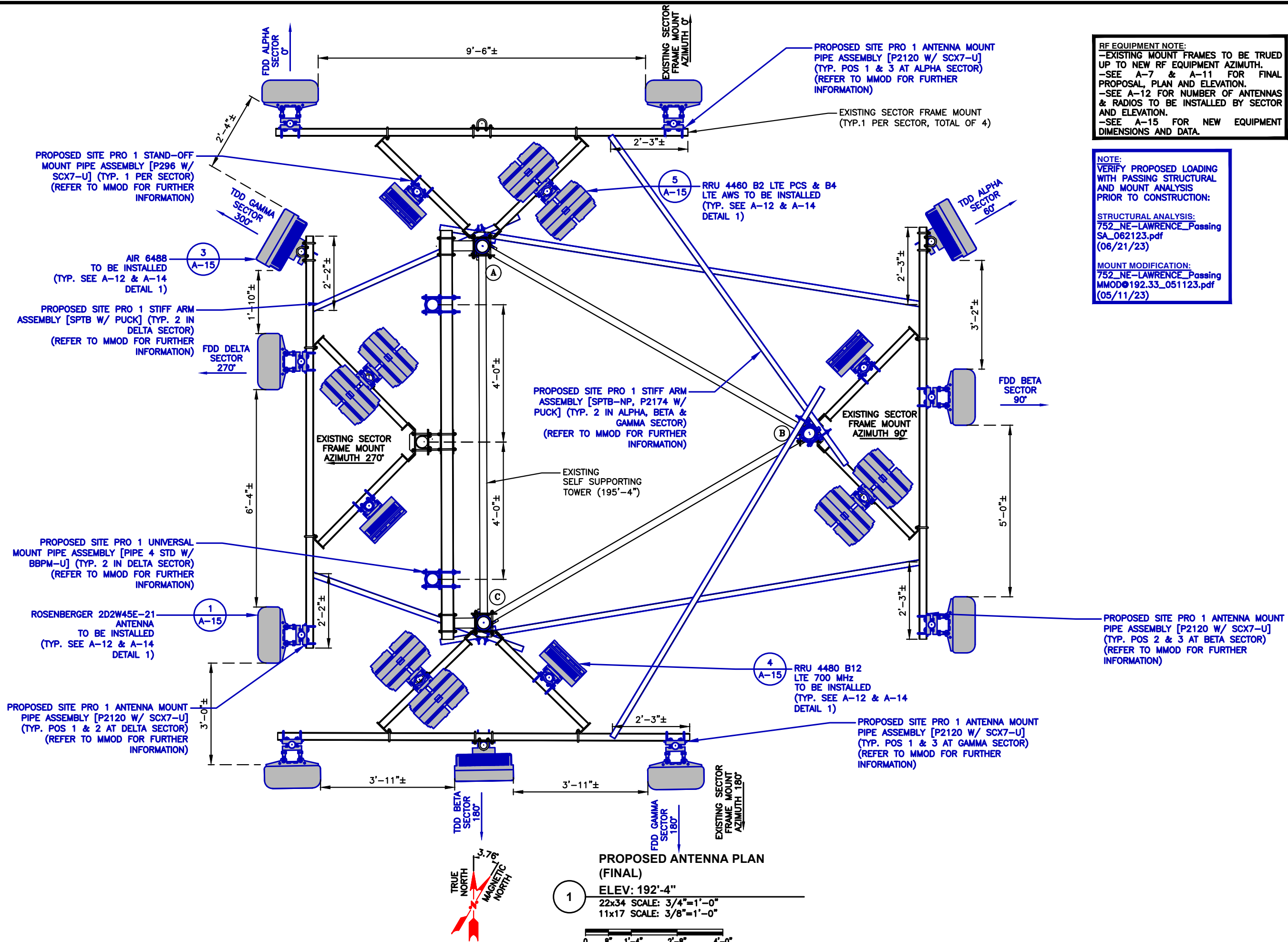
SHEET TITLE

PROPOSED ANTENNA PLAN
VIEW (TEMPORARY) (HOT
CUT)

SHEET NUMBER

A-6

752_NE-LAWRENCE_REVO_062823.DWG



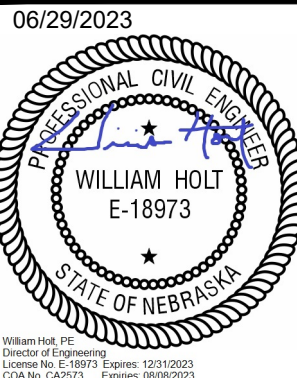
VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701



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NE-LAWRENCE

SITE NUMBER:
752

SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE

PROPOSED ANTENNA
PLAN VIEW (FINAL)
ELEV: 192'-4"

SHEET NUMBER

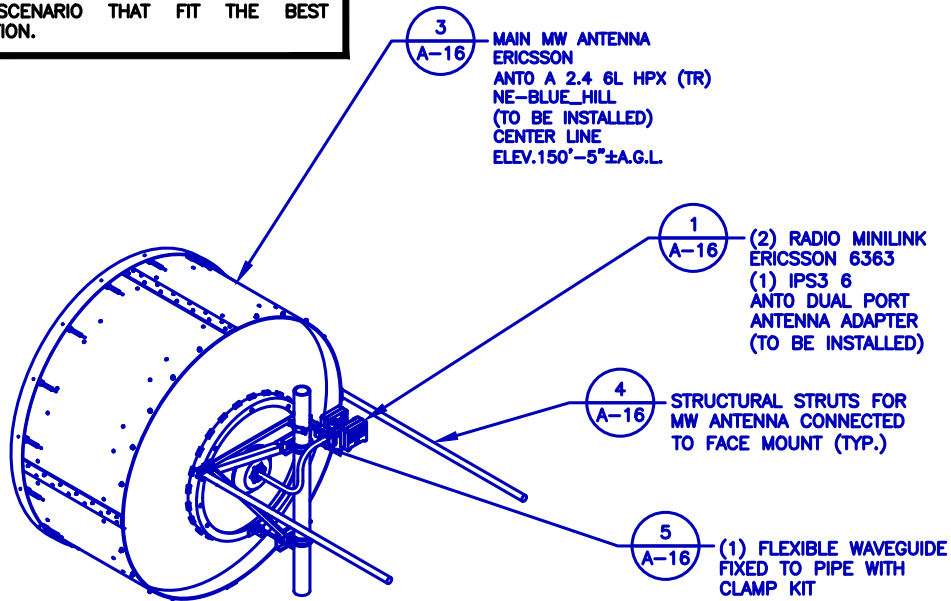
A-7

NOTE:
SEE A-16 FOR NEW MW
EQUIPMENT DIMENSIONS
AND DATA

STRUTS NOTE:
STIFF ARMS AND
FASTENERS TO BE
INSTALLED PER
MANUFACTURER'S
SPECIFICATIONS

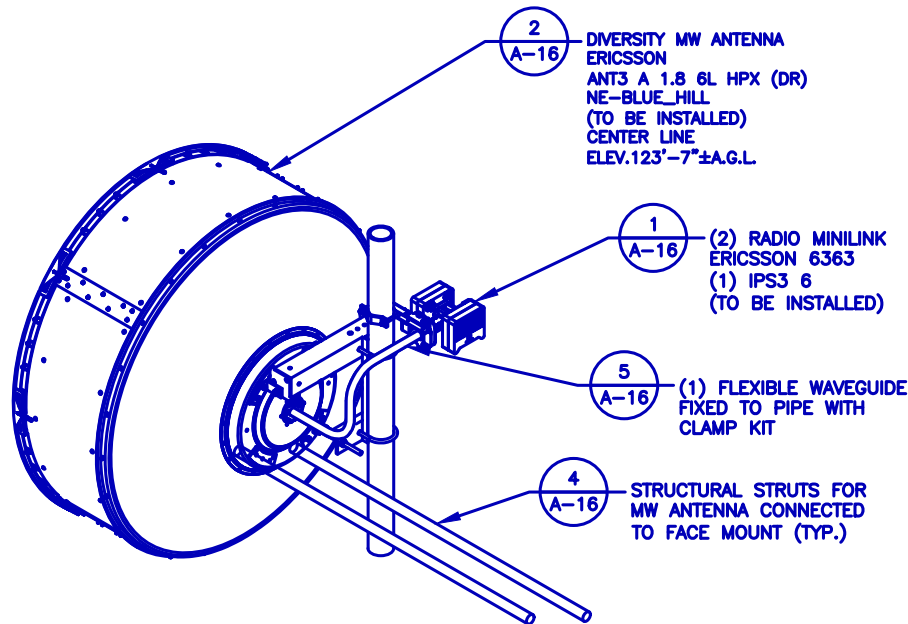
NEW FACE MOUNT NOTE:
PROPOSED FACE MOUNT
MODEL CAN CHANGE BASED
IN AVAILABILITY ON-HAND
OR WHAT IS IN-STOCK
WITH THE SUPPLIER.

NOTE:
-FOR MW MIGRATION DETAILS REFER TO
ALL TEDs APPLICABLE PER LINK AND TO
"MICROWAVE MW RIP AND REPLACE
SEQUENCING" FILE.
-THE MIGRATION WOULD BE DEPENDANT
ON SCENARIO THAT FIT THE BEST
SOLUTION.



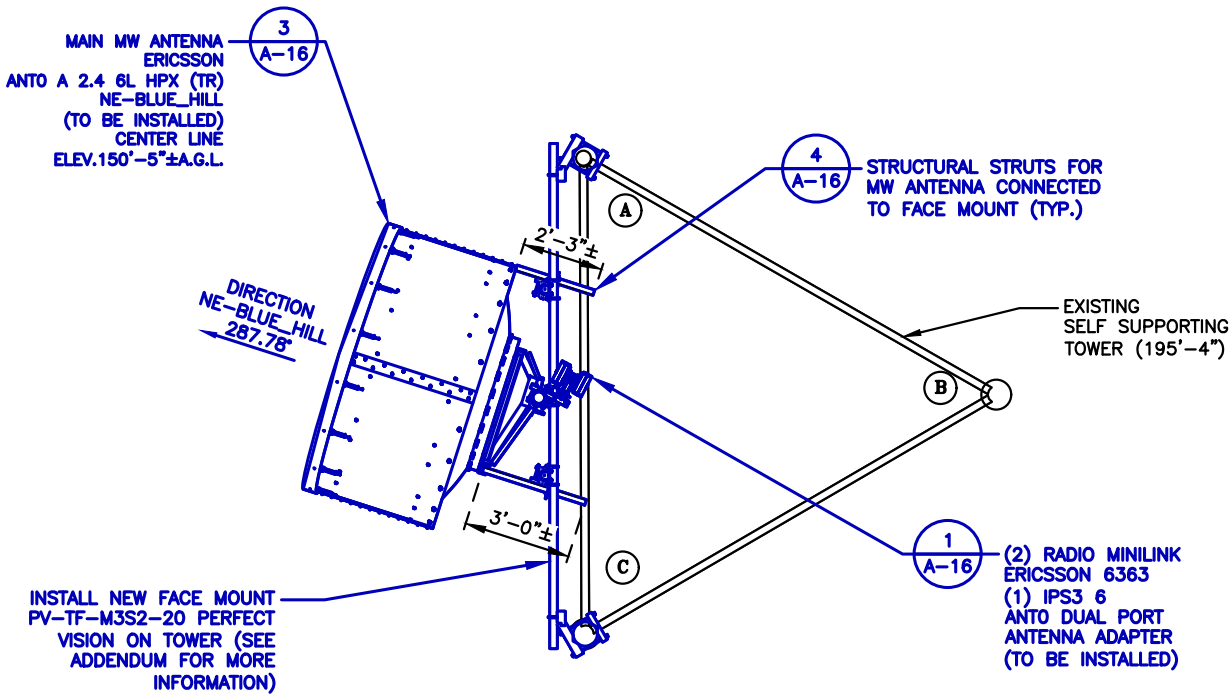
PROPOSED MAIN MW ANTENNA
ANTO A 2.4 6L HPX (TR) DETAIL

ELEV: 150'-5"
SCALE: N.T.S



PROPOSED DIVERSITY MW ANTENNA
ANT3 A 1.8 6L HPX (DR) DETAIL

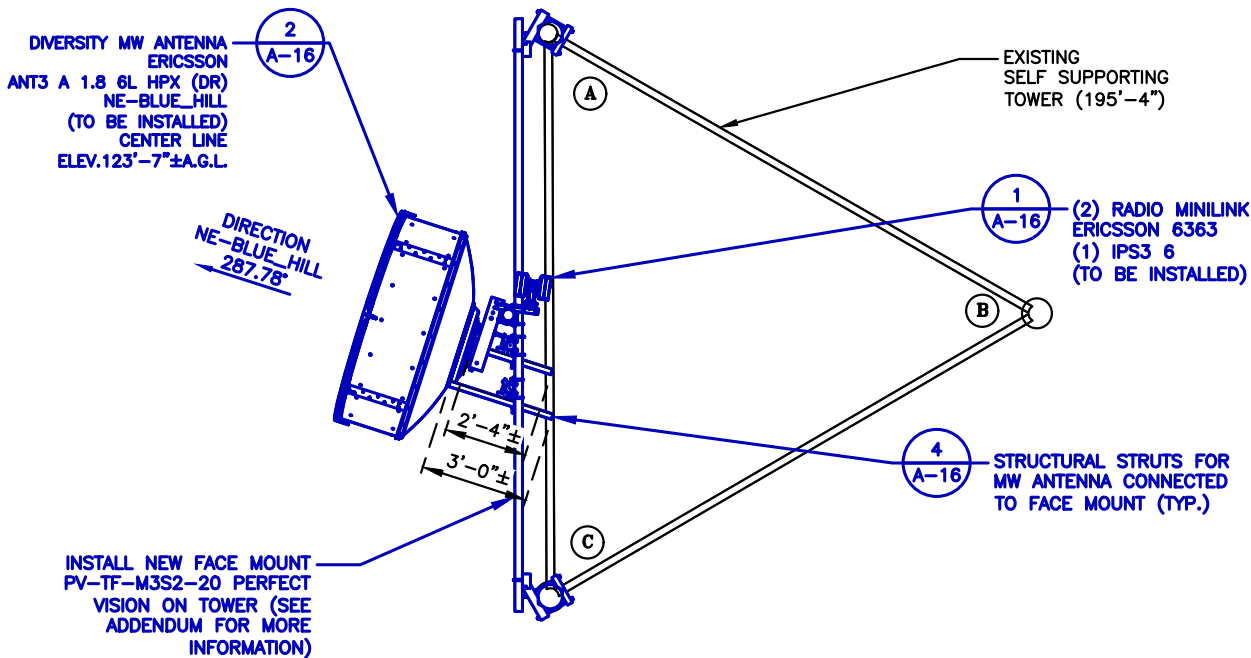
ELEV: 123'-7"
SCALE: N.T.S



PROPOSED MW ANTENNA PLAN

ELEV: 150'-5"

22x34 SCALE: 3/8"=1'-0"
11x17 SCALE: 3/16"=1'-0"



PROPOSED MW ANTENNA PLAN

ELEV: 123'-7"

22x34 SCALE: 3/8"=1'-0"
11x17 SCALE: 3/16"=1'-0"



VIAERO
WIRELESS

VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

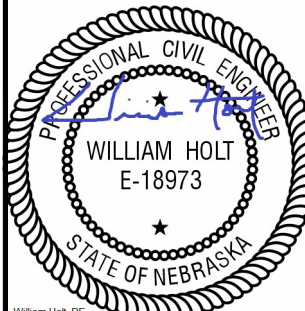
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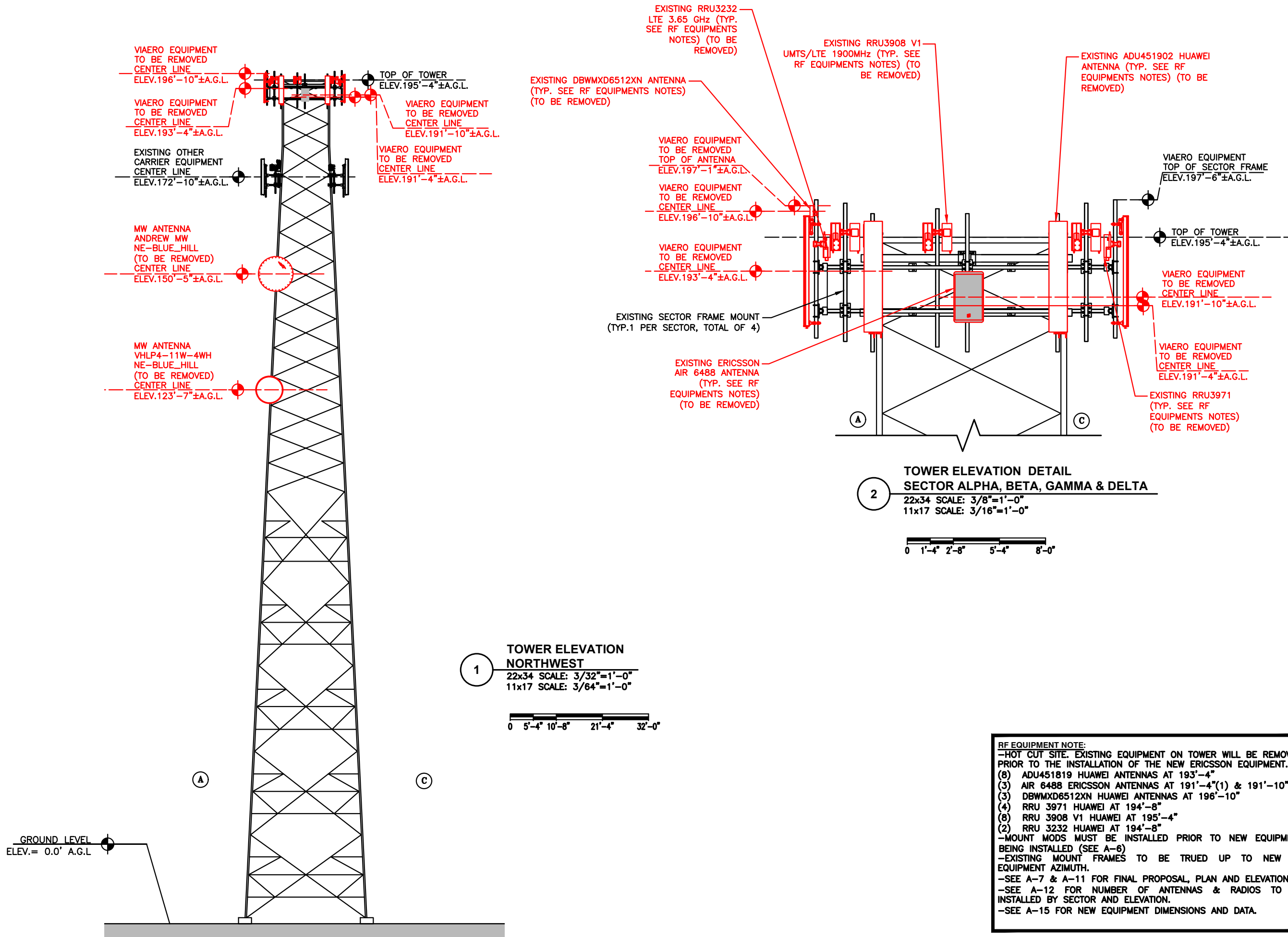
SHEET TITLE

PROPOSED MW
ANTENNA PLAN ELEV

SHEET NUMBER

A-8

752_NE-LAWRENCE_REVO_062823.DWG



VIAERO
WIRELESS

VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

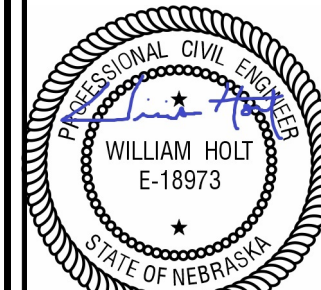
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SITE NAME:
NE-LAWRENCE

SITE NUMBER:
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SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE

EXISTING ELEVATIONS

SHEET NUMBER

A-9

752_NE-LAWRENCE_REVO_062823.DWG

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(3) AIR 6488 ERICSSON ANTENNAS AT 191'-4"(1) & 191'-10"(2)
(3) DBWMXD6512XN HUAWEI ANTENNAS AT 196'-10"
(4) RRU 3971 HUAWEI AT 194'-8"
(8) RRU 3908 V1 HUAWEI AT 195'-4"
(2) RRU 3232 HUAWEI AT 194'-8"
-MOUNT MODS MUST BE INSTALLED PRIOR TO NEW EQUIPMENT
BEING INSTALLED (SEE A-6)
-EXISTING MOUNT FRAMES TO BE TRUED UP TO NEW RF
EQUIPMENT AZIMUTH.
-SEE A-7 & A-11 FOR FINAL PROPOSAL, PLAN AND ELEVATION.
-SEE A-12 FOR NUMBER OF ANTENNAS & RADIOS TO BE
INSTALLED BY SECTOR AND ELEVATION.
-SEE A-15 FOR NEW EQUIPMENT DIMENSIONS AND DATA.



VIAERO WIRELESS
1224 WEST PLATTE AVENUE FORT MORGAN, CO 80701

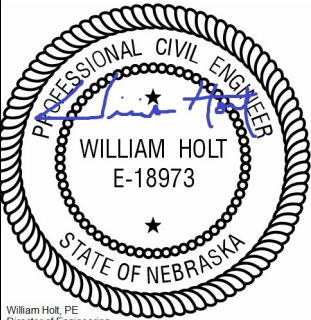


6300 LEGACY DRIVE
PLANO, TX 75024



319 CHAPMANKE RD, SUITE 118
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PH: (405)348-5480 FAX: (405)341-4825

06/29/2023



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Director of Engineering
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COA No. CA2573 Expires: 08/08/2023

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O	06/28/23	FOR CONSTRUCTION	RAS

DRAWING BY: RAS CHECKED BY: OBN

SITE NAME:
NE-LAWRENCE

SITE NUMBER:
752

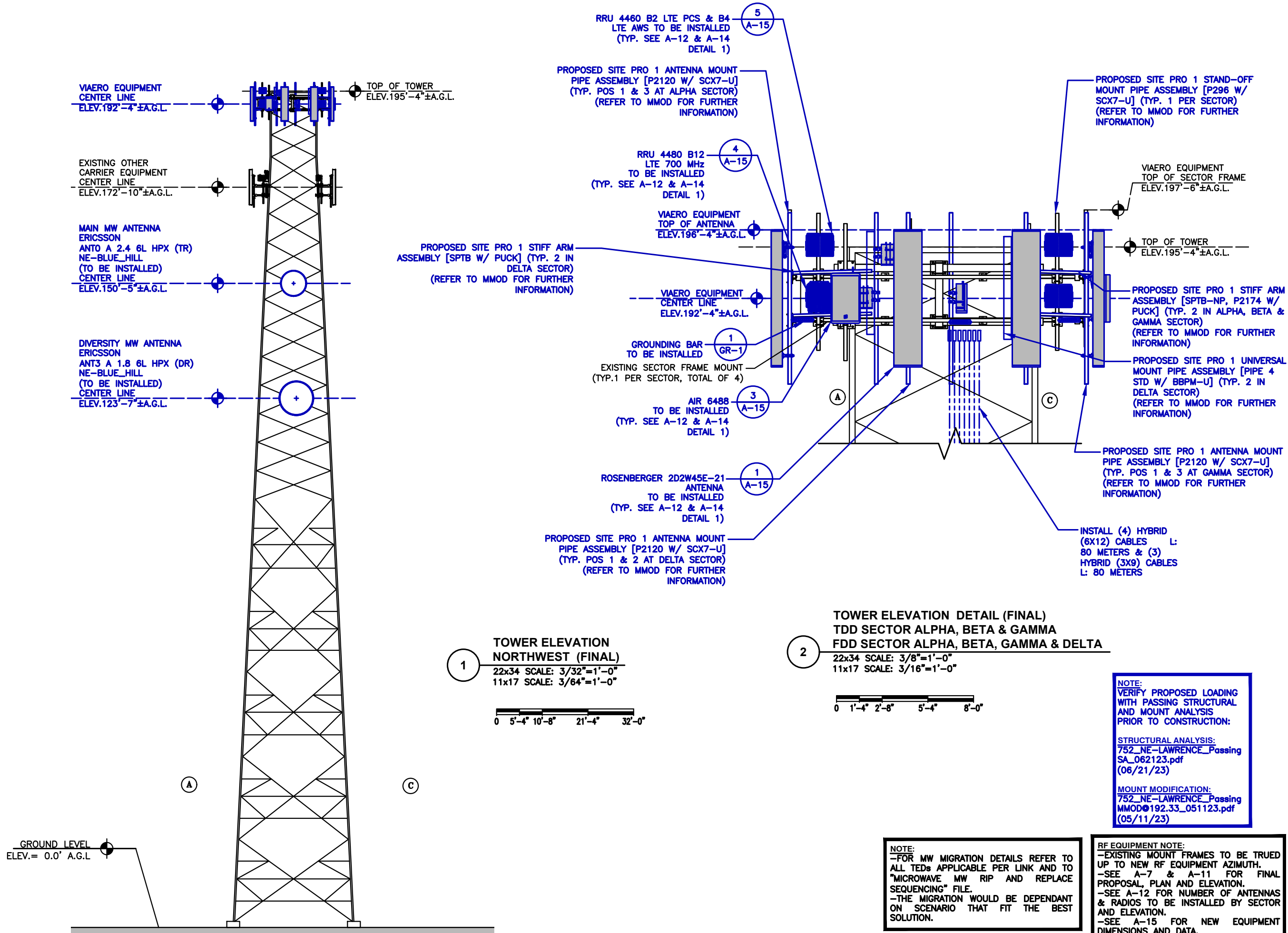
SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE
PROPOSED
ELEVATIONS
(TEMPORARY) (HOT
CUT)

SHEET NUMBER

A-10

752_NE-LAWRENCE_REVO_062823.DWG



VIAERO
WIRELESS

VIAERO WIRELESS
1224 WEST PLATE AVENUE FORT MORGAN, CO 80701

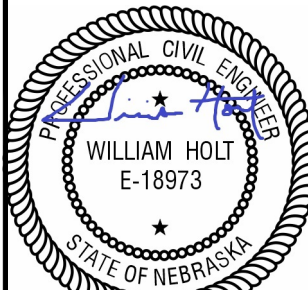
ERICSSON

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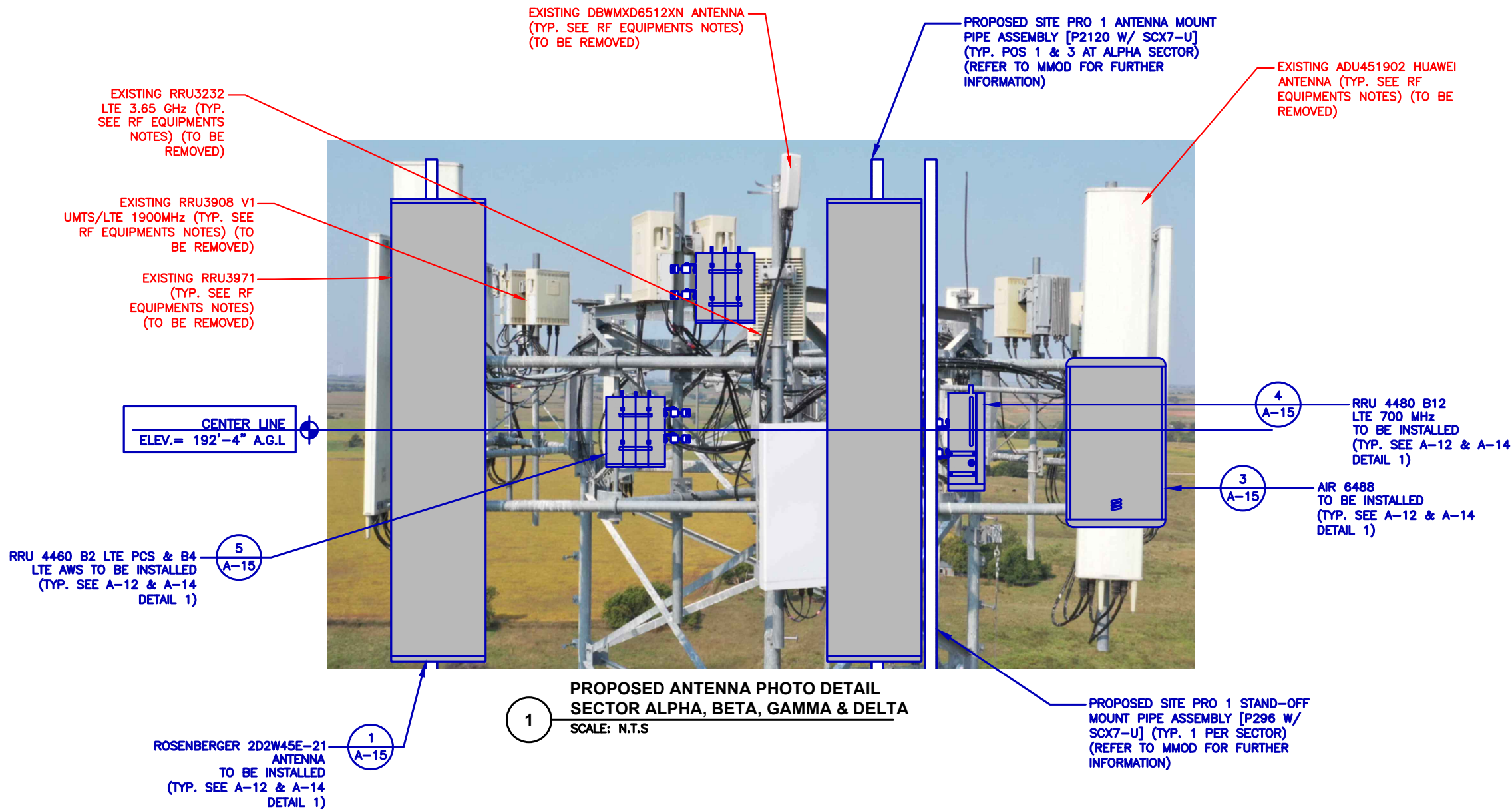
SHEET TITLE

PROPOSED
ELEVATIONS
(FINAL)

SHEET NUMBER

A-11

752_NE-LAWRENCE_REV0_062823.DWG



RF EQUIPMENT NOTE:
-HOT CUT SITE. EXISTING EQUIPMENT ON TOWER WILL BE REMOVED PRIOR TO THE INSTALLATION OF THE NEW ERICSSON EQUIPMENT.
(8) ADU451819 HUAWEI ANTENNAS AT 193'-4"
(3) AIR 6488 ERICSSON ANTENNAS AT 191'-4"(1) & 191'-10"(2)
(3) DBWMXD6512XN HUAWEI ANTENNAS AT 196'-10"
(4) RRU 3971 HUAWEI AT 194'-8"
(8) RRU 3908 V1 HUAWEI AT 195'-4"
(2) RRU 3232 HUAWEI AT 194'-8"
-MOUNT MODS MUST BE INSTALLED PRIOR TO NEW EQUIPMENT BEING INSTALLED (SEE A-6)
-EXISTING MOUNT FRAMES TO BE TRUED UP TO NEW RF EQUIPMENT AZIMUTH.
-SEE A-7 & A-11 FOR FINAL PROPOSAL, PLAN AND ELEVATION.
-SEE A-12 FOR NUMBER OF ANTENNAS & RADIOS TO BE INSTALLED BY SECTOR AND ELEVATION.
-SEE A-15 FOR NEW EQUIPMENT DIMENSIONS AND DATA.

RF SYSTEM SCHEDULE								
SECTOR	ANTENNA CENTERLINE	AZIMUTH	VENDOR	ANTENNA MODEL	FREQUENCY / TECHNOLOGY	MECH. TILT	RADIO MODEL	CABLE TYPE & LENGTH
TDD ALPHA	192'-4"	60°	ERICSSON	AIR 6488	CBRS	0°	INTEGRATED RADIO	(1) 3x9 HYBRID CABLE 80 METERS
FDD ALPHA	192'-4"	0°	ERICSSON	ROSENBERGER 2D2W45E-21	700 MHz / LTE 1900 MHz PCS / LTE 2100 MHz AWS / LTE	0°	(1) ERICSSON RRU 4480 (2) ERICSSON RRU 4460	(1) 6X12 HYBRID CABLE 80 METERS
FDD ALPHA	192'-4"	0°	ERICSSON	ROSENBERGER 2D2W45E-21		0°		
TDD BETA	192'-4"	180°	ERICSSON	AIR 6488	CBRS	0°	INTEGRATED RADIO	(1) 3x9 HYBRID CABLE 80 METERS
FDD BETA	192'-4"	90°	ERICSSON	ROSENBERGER 2D2W45E-21	700 MHz / LTE 1900 MHz PCS / LTE 2100 MHz AWS / LTE	0°	(1) ERICSSON RRU 4480 (2) ERICSSON RRU 4460	(1) 6X12 HYBRID CABLE 80 METERS
FDD BETA	192'-4"	90°	ERICSSON	ROSENBERGER 2D2W45E-21		0°		
TDD GAMMA	192'-4"	300°	ERICSSON	AIR 6488	CBRS	0°	INTEGRATED RADIO	(1) 3x9 HYBRID CABLE 80 METERS
FDD GAMMA	192'-4"	180°	ERICSSON	ROSENBERGER 2D2W45E-21	700 MHz / LTE 1900 MHz PCS / LTE 2100 MHz AWS / LTE	0°	(1) ERICSSON RRU 4480 (2) ERICSSON RRU 4460	(1) 6X12 HYBRID CABLE 80 METERS
FDD GAMMA	192'-4"	180°	ERICSSON	ROSENBERGER 2D2W45E-21		0°		
FDD DELTA	192'-4"	270°	ERICSSON	ROSENBERGER 2D2W45E-21	700 MHz / LTE 1900 MHz PCS / LTE 2100 MHz AWS / LTE	0°	(1) ERICSSON RRU 4480 (2) ERICSSON RRU 4460	(1) 6X12 HYBRID CABLE 80 METERS
FDD DELTA	192'-4"	270°	ERICSSON	ROSENBERGER 2D2W45E-21		0°		

TOTAL OF NEW HYBRID CABLES:
INSTALL (4) HYBRID (6X12) CABLES L: 80 METERS &
(3) HYBRID (3X9) CABLES L: 80 METERS

VIAERO
WIRELESS

VIAERO WIRELESS
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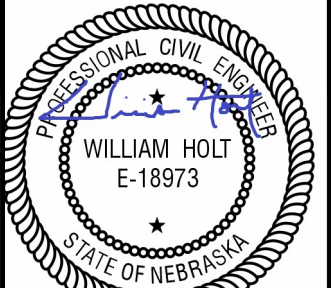
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LAWRENCE, NE 68957

SHEET TITLE

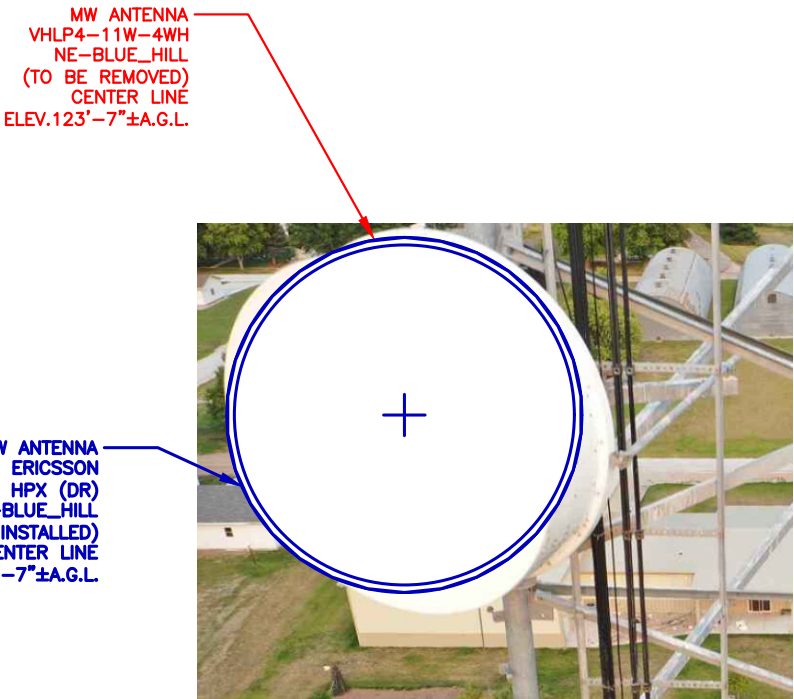
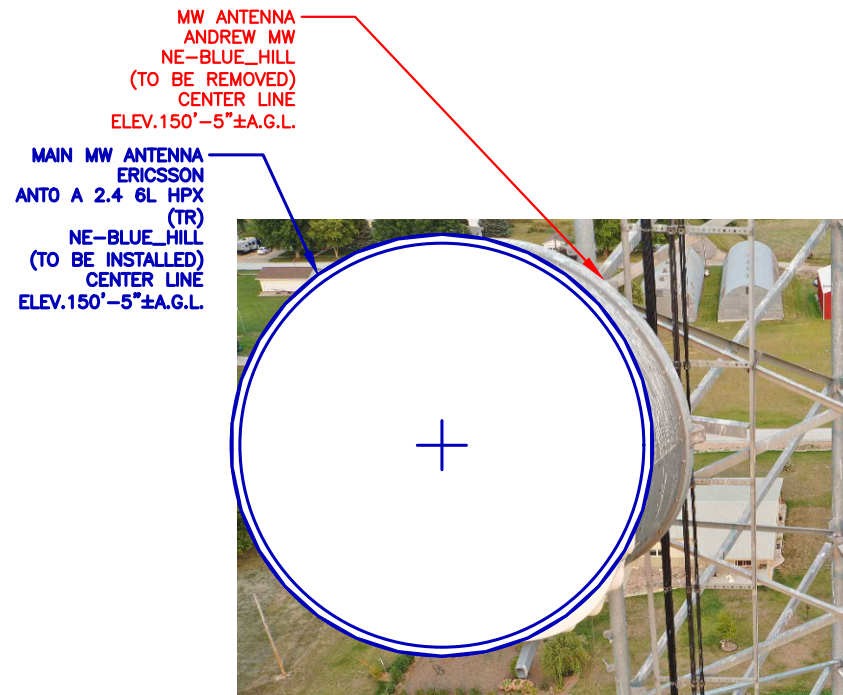
RF SYSTEM
SCHEDULE
(FINAL)

SHEET NUMBER

A-12

NOTE:
SEE A-16 FOR NEW MW
EQUIPMENT DIMENSIONS
AND DATA

NOTE:
-FOR MW MIGRATION DETAILS REFER TO
ALL TEDs APPLICABLE PER LINK AND TO
"MICROWAVE MW RIP AND REPLACE
SEQUENCING" FILE.
-THE MIGRATION WOULD BE DEPENDANT
ON SCENARIO THAT FIT THE BEST
SOLUTION.



MW SYSTEM SCHEDULE									
	ANTENNA CENTERLINE	AZIMUTH	DIRECTION	VENDOR	ANTENNA MODEL	LEG	CONFIGURATION	RADIO UNIT	CABLE TYPE AND LENGTH
MAIN	150'-5"	287.78°	NE-BLUE_HILL	ERICSSON	ANTO A 2.4 6L HPX (TR)	A-C	(2+2 SD ACAP)	(2) MINILINK 6363	(2) COAX. CABLES 195 FT.
DIVERSITY	123'-7"	287.78°	NE-BLUE_HILL	ERICSSON	ANT3 A 1.8 6L HPX (DR)	A-C	(2+2 SD ACAP)	(2) MINILINK 6363	(2) COAX. CABLES 165 FT.

VIAERO
WIRELESS

VIAERO WIRELESS
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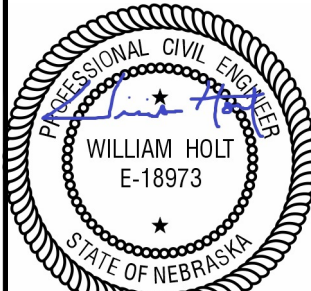
ERICSSON

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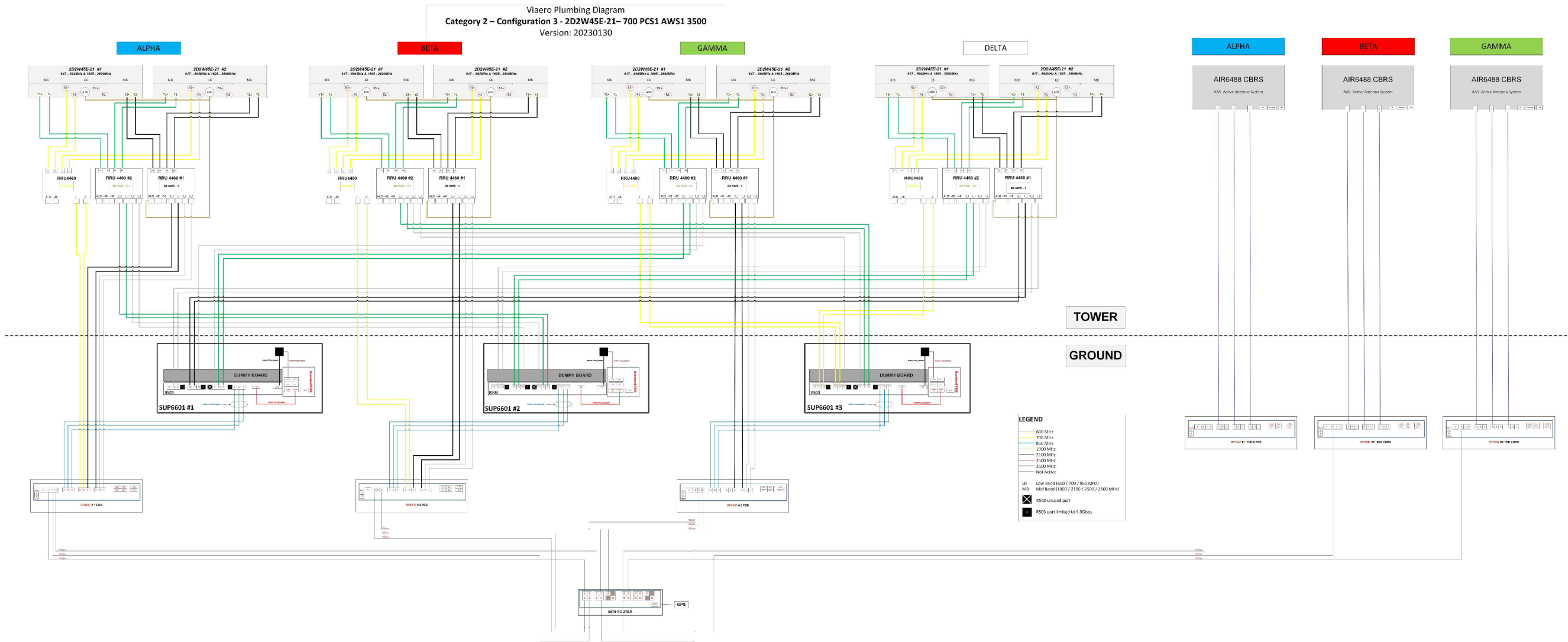
SHEET TITLE

MW SYSTEM
SCHEDULE

SHEET NUMBER

A-13

752_NE-LAWRENCE_REV0_062823.DWG



1 RF PLUMBING DIAGRAM
SCALE: N.T.S

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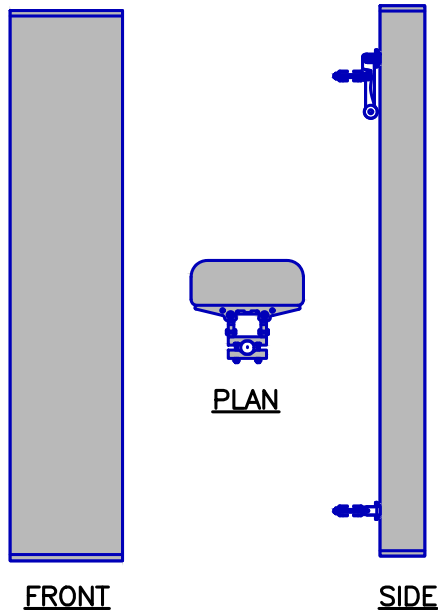
SHEET TITLE

PLUMBING DIAGRAMS

SHEET NUMBER

A-14

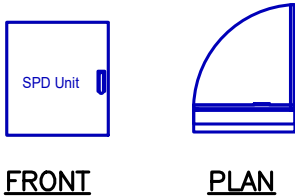
752_NE-LAWRENCE_REV0_062823.DWG



ANTENNA DIMENSIONS	
MODEL #	ROSENBERGER_2D2W45E-21
MANUF.	ROSENBERGER
HEIGHT	96"
WIDTH	19.6"
DEPTH	7.8"
WEIGHT	82 lbs
CONNECTION TYPE	12 x 4.3-10 female

1

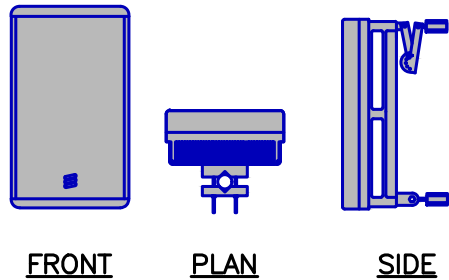
ROSENBERGER ANTENNA DETAIL
SCALE: N.T.S



SPD DIMENSIONS	
MODEL #	BMG907052/6
MANUF.	ERICSSON
WIDTH	17.70"
DEPTH	4.70"
HEIGHT	19.70"
WEIGHT	39.70 LBS
(INSTALL PER MANUFACTURER'S INSTALLATION GUIDELINES)	

2

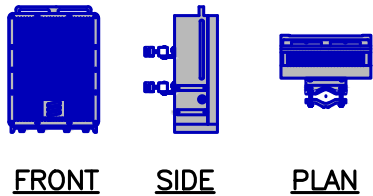
SPD BOX DETAIL
SCALE: N.T.S



ANTENNA DIMENSIONS	
MODEL #	AIR 6488
MANUF.	ERICSSON
HEIGHT	35.1"
WIDTH	20.5"
DEPTH	9.4"
WEIGHT	135 LBS

3

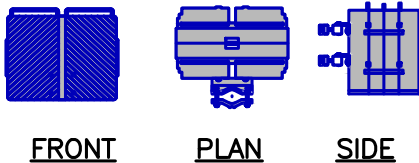
AIR 6488 DETAIL
SCALE: N.T.S



RADIO DIMENSIONS	
MODEL #	RRU 4480
MANUF.	ERICSSON
HEIGHT	21.92"
WIDTH	15.66"
DEPTH	7.48"
WEIGHT	80.90 LBS
CONNECTION TYPE	4 x 4.3-10 FEMALE

4

RADIO 4480 DETAIL
SCALE: N.T.S



RADIO DIMENSIONS	
MODEL #	RADIO 4460
MANUF.	ERICSSON
HEIGHT	19.72"
WIDTH	15.78"
DEPTH	12.20"
WEIGHT	109.12 LBS

5

RADIO 4460 DETAIL
SCALE: N.T.S

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WIRELESS

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ERICSSON

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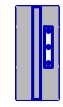
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150 S. SMITH STREET
LAWRENCE, NE 68507

SHEET TITLE

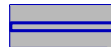
RF EQUIPMENT
DETAILS

SHEET NUMBER

A-15



SIDE



PLAN



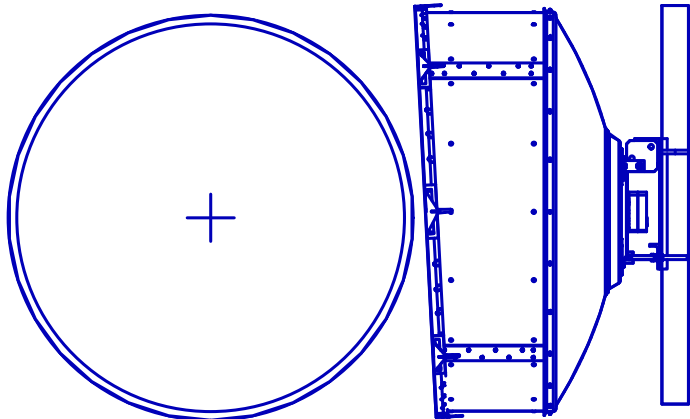
FRONT

RADIO LINK DIMENSIONS

MODEL #	ML 6363
MANUF.	ERICSSON
HEIGHT	7"
WIDTH	7.8"
DEPTH	3.1"
WEIGHT	6.20 LBS

1

ML6363 DETAIL
SCALE: N.T.S



FRONT

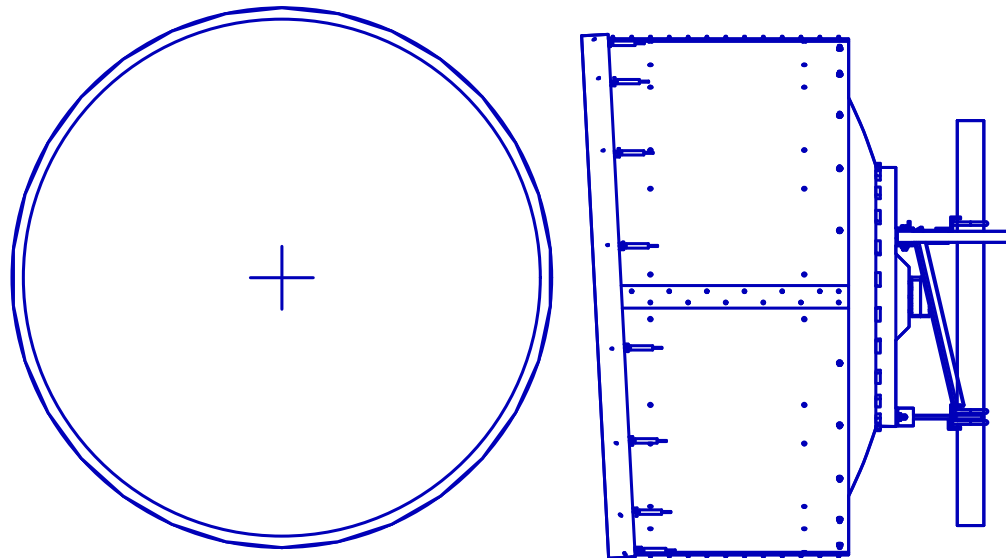
SIDE

MW ANTENNA DIMENSIONS

MODEL #	ANTO A 1.8 / ANT3 A 1.8
MANUF.	ERICSSON
DIAMETER	6'
HEIGHT	6.23'
WIDTH	6.23'
DEPTH	3.36'
WEIGHT	207.25 lbs
FREQUENCY	6 GHz

2

6FT MW ANTENNA DETAIL
SCALE: N.T.S



FRONT

SIDE

MW ANTENNA DIMENSIONS

MODEL #	ANTO A 2.4 / ANT3 A 2.4
MANUF.	ERICSSON
DIAMETER	8'
HEIGHT	7.92'
WIDTH	7.92'
DEPTH	5.51'
WEIGHT	412.26 lbs
FREQUENCY	6 GHz

3

8FT MW ANTENNA DETAIL
SCALE: N.T.S



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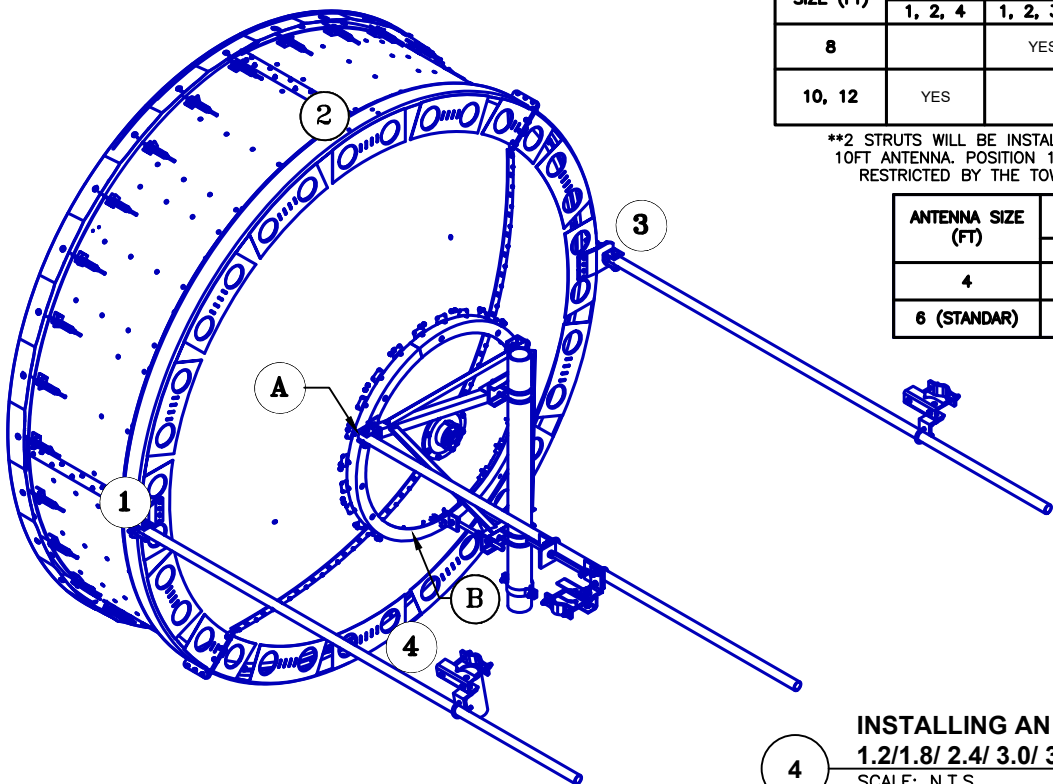
SITE ADDRESS:
150 S. SMITH STREET
LAWRENCE, NE 68957

SHEET TITLE

MW EQUIPMENT
DETAILS

SHEET NUMBER

A-16



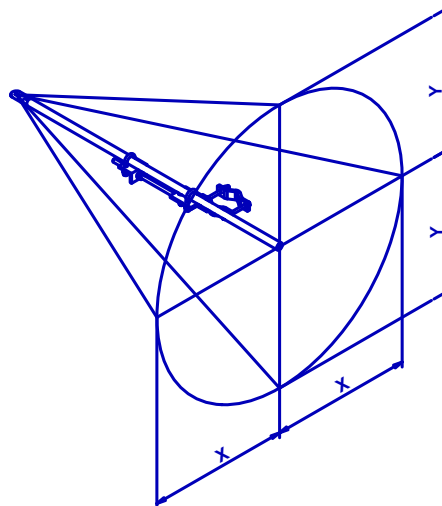
4

INSTALLING AN ANT SIDE STRUT
1.2/1.8/ 2.4/ 3.0/ 3.7
SCALE: N.T.S

ANTENNA SIZE (FT)	POSITION**		
	1, 2, 4	1, 2, 3, 4	3
8		YES	
10, 12	YES		SUPPLIED AS A STANDARD WITH ANTENNA

**2 STRUTS WILL BE INSTALLED PER 8FT &
10FT ANTENNA. POSITION 1 TO 4 WILL BE
RESTRICTED BY THE TOWER ON SITE

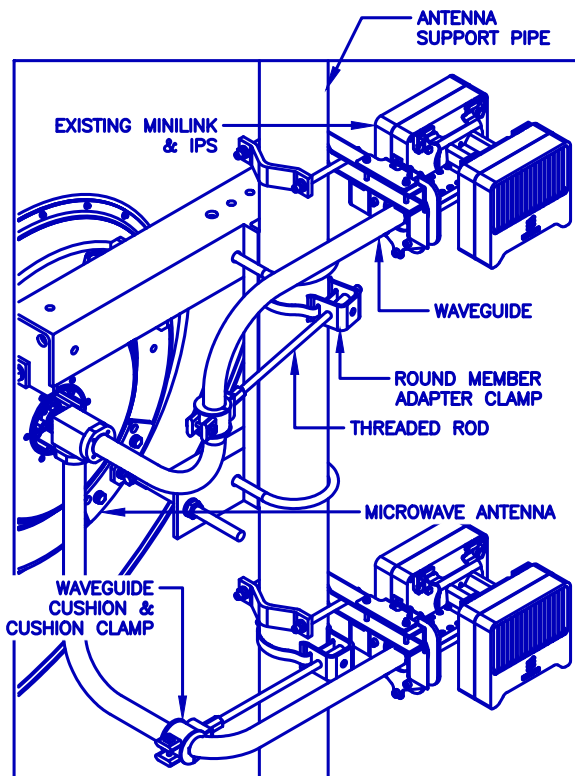
ANTENNA SIZE (FT)	
	A,B
4	YES
6 (STANDAR)	YES



LIMITS OF STRUT*		
ANTENNA SIZE (FT)	X	Y
4, 6	12"	15"
8, 10	25"	25"

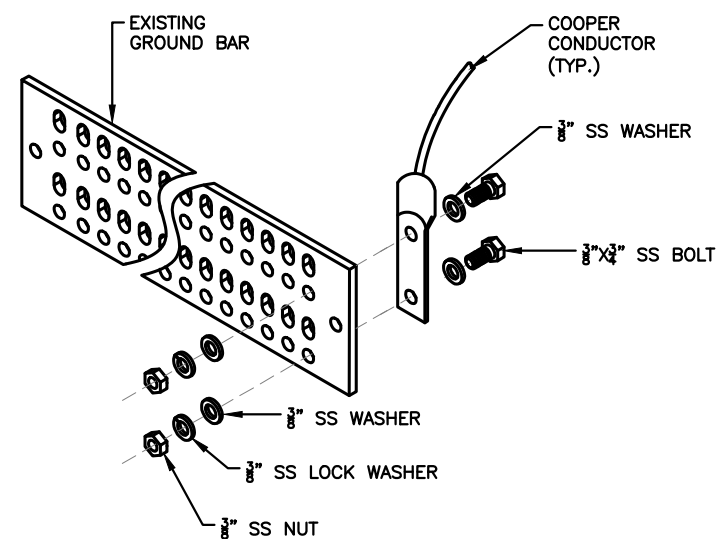
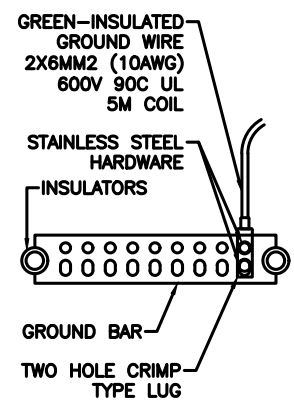
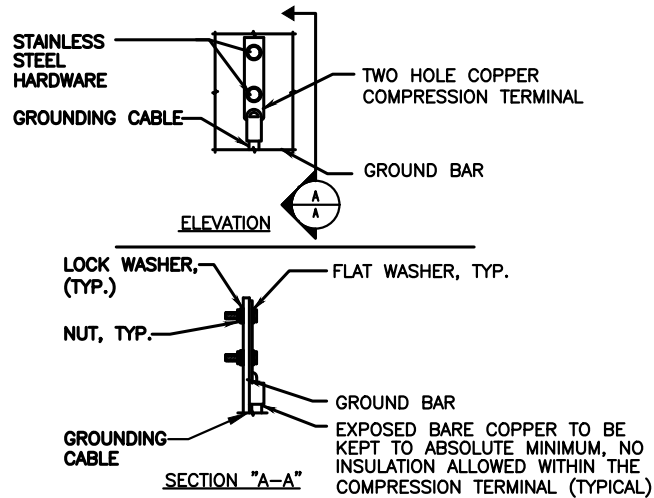
STRUT LENGTHS MIN*	
STRUT	LENGTH
ANT SIDE STRUT A 0.9	1550mm/ 61"
ANT SIDE STRUT A 1.2/1.8	1550mm/ 61"
ANT SIDE STRUT A 2.4/3.0/3.7	3000mm/ 118"

*ALL SPECIFICATIONS ACCORDING TO MANUFACTURER
RECOMMENDATIONS. ANGLE MAY LIMITED BY DIRECT
MOUNT RADIO EQUIPMENT.



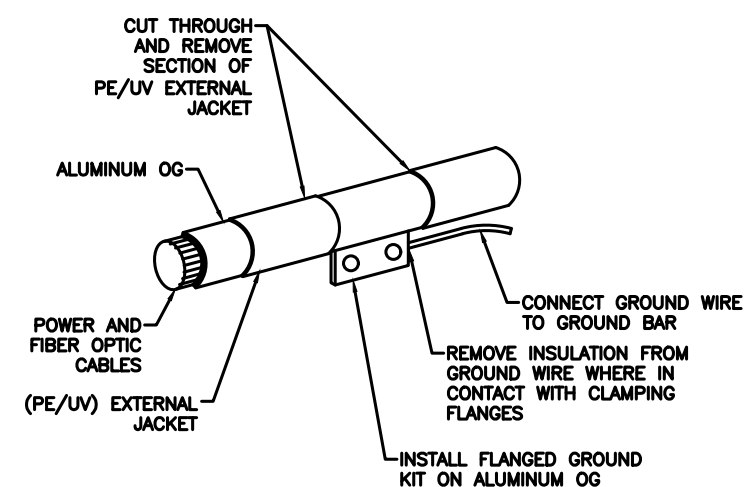
5

WAVEGUIDE CLAMP DETAIL
SCALE: N.T.S

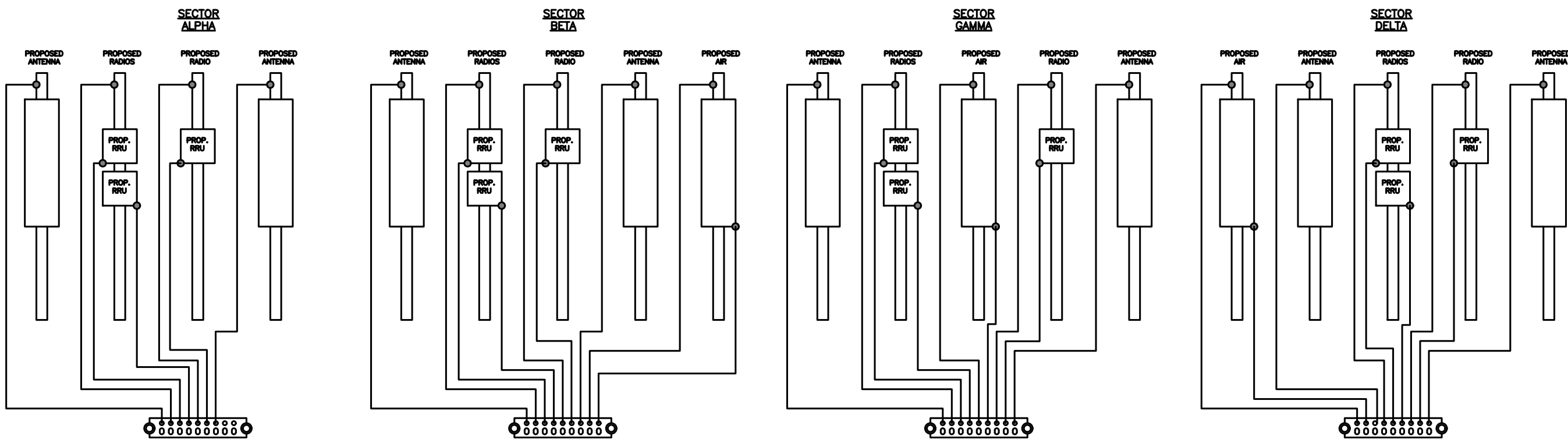


NOTES:
ALL EXPOSED COPPER-BASED METALS ARE REQUIRED TO BE COMPLETELY COVERED WITH WATERPROOF MATERIALS TO PREVENT COPPER ION EROSION OF STEEL TANK STRUCTURE.

1 GROUNDING BAR DETAILS
SCALE: N.T.S



2 HYBRID CABLE GROUNDING DETAIL
SCALE: N.T.S



3 ANTENNA & RADIOS GROUNDING DIAGRAM
SCALE: N.T.S



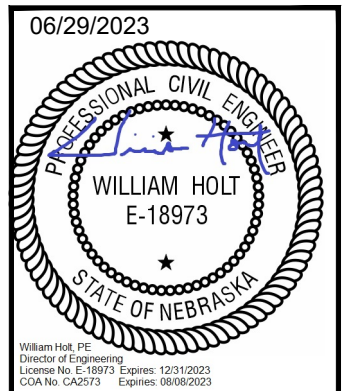
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LAWRENCE, NE 68957

SHEET TITLE

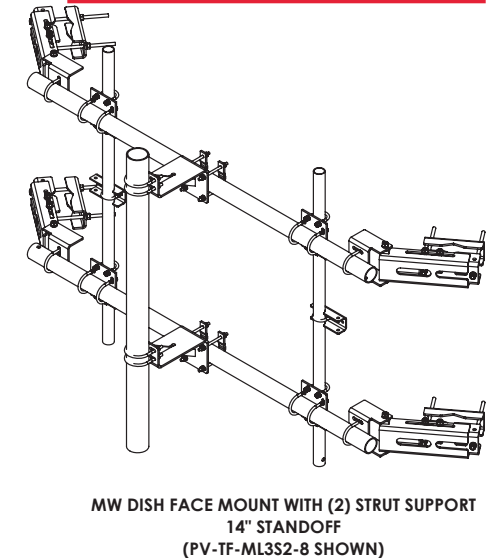
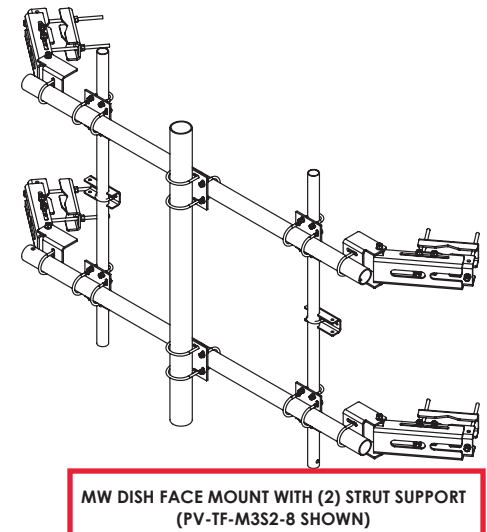
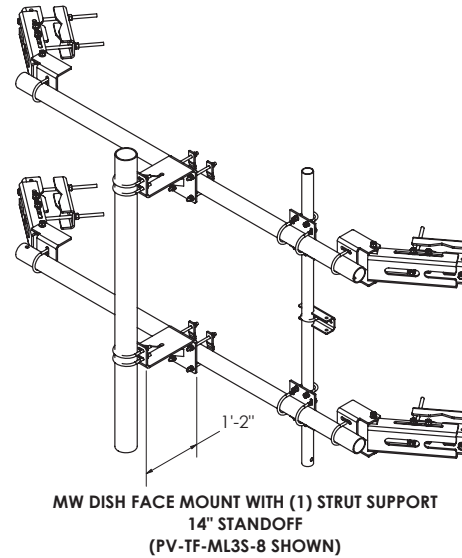
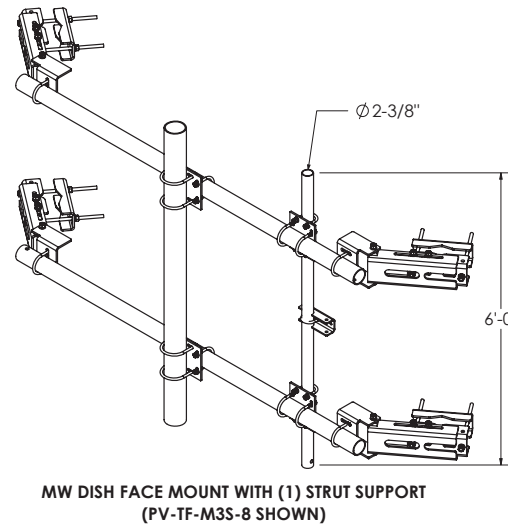
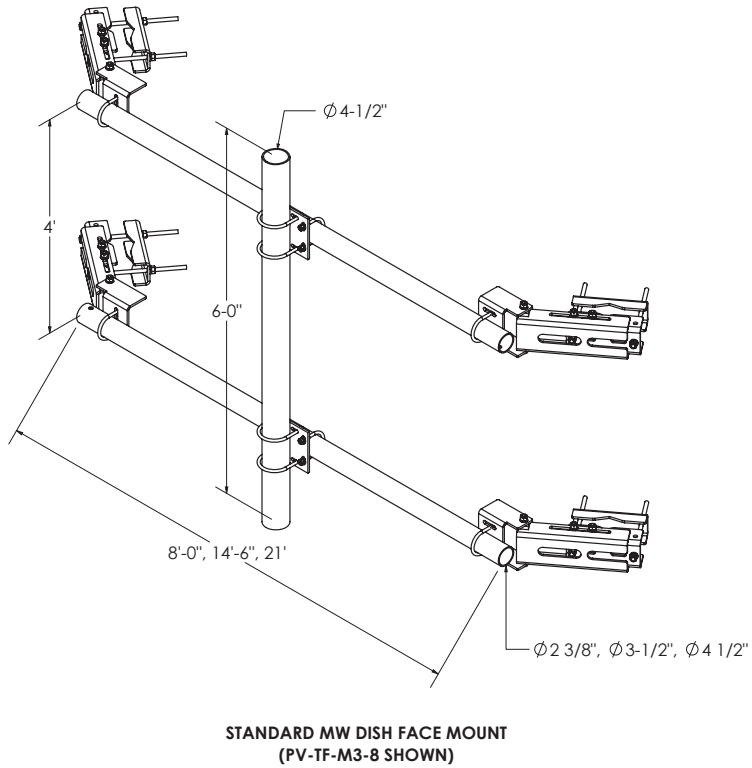
GROUNDING DETAILS

SHEET NUMBER

GR-1

PV-TF-M

TOWER FACE MOUNT



Series	Description	Compatible 60 deg Angle(in)	Compatible 90 deg Angle(in)	Compatible Round Leg (in)
PV-TF-M	Standard Leg	Up to 8x8	Up to 6x6	Up to 8.625 OD
PV-TFL-M	Large Leg	Up to 16x16	Up to 12x12	Up to 16 OD

SHEET	THIRD ANGLE PROJECTION	CATEGORY	01_Lattice Tower	4		
1 OF 7	SCALE 1:1	SERIES	06_Face Mounts	3		
2/9/2021		TYPE	PV-TF_Tower Face Mount	2	PV-TF/TFL-MD REPLACED BY PV-TF/TFL-MB, PV-CO-U REPLACED BY PV-SMART-MSK4545	1/28/21
		BY	INT	1	TEMPLATE UPDATE	11/27/19
		CHECKED	DJN	0	INITIAL RELEASE	1/13/16
		STATUS	APPROVED	REV	DESCRIPTION	DATE
DIMENSIONS ARE IN INCHES TOLERANCES U.N.O. HOLES: +1/16", -1/32" ANGULAR: PROFILE ±1/4", BEND ±2° ALL OTHERS: ±1/16"						
PERFECT VISION TOWER FACE MOUNT DOCUMENT NUMBER TF-ENG-01-R2						REV 2

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF PERFECTVISION MFG. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF PERFECTVISION MFG. IS PROHIBITED.

	Part Number	Description	Weight (lbs)	Included Parts																
				Leg Bracket	Pipe										Crossover / Standoff			U-Bolts		
PV-TF-MB	PV-TFL-MB	PV-MDL-CS	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	PV-TF-MB	
PV-TF-M#-#	Standard MW Dish Face Mount																			
PV-TF-M2-8	2-3/8" x 8" Face	388	1	-	-	-	2	-	-	-	-	-	-	-	-	2	-	-	4	
PV-TF-M2-14	2-3/8" x 14" Face	435	1	-	-	-	2	-	-	-	-	-	-	-	-	2	-	-	4	
PV-TF-M2-20	2-3/8" x 21" Face	483	1	-	-	-	-	2	-	-	-	-	-	-	-	2	-	-	4	
PV-TF-M3-8	3-1/2" x 8" Face	453	1	-	-	-	-	2	-	-	-	-	-	-	-	2	-	-	4	
PV-TF-M3-14	3-1/2" x 14" Face	551	1	-	-	-	-	-	2	-	-	-	-	-	-	2	-	-	4	
PV-TF-M3-20	3-1/2" x 21" Face	650	1	-	-	-	-	-	-	2	-	-	-	-	-	2	-	-	4	
PV-TF-M4-8	4-1/2" x 8" Face	510	1	-	-	-	-	-	-	-	1	2	-	-	-	-	2	-	4	
PV-TF-M4-14	4-1/2" x 14" Face	640	1	-	-	-	-	-	-	-	-	1	2	-	-	-	2	-	4	
PV-TF-M4-20	4-1/2" x 21" Face	792	1	-	-	-	-	-	-	-	-	1	-	2	-	-	2	-	4	
PV-TF-M#S-#	MW Dish Face Mount with (1) Strut Support																			
PV-TF-M2S-8	2-3/8" x 8" Face, 1 Strut	436	1	-	-	1	2	-	-	-	-	1	-	-	-	2	-	1	4	
PV-TF-M2S-14	2-3/8" x 14" Face, 1 Strut	483	1	-	-	1	2	-	-	-	-	1	-	-	-	2	-	1	4	
PV-TF-M2S-20	2-3/8" x 21" Face, 1 Strut	531	1	-	-	1	-	2	-	-	-	1	-	-	-	2	-	1	4	
PV-TF-M3S-8	3-1/2" x 8" Face, 1 Strut	501	1	-	-	1	-	-	2	-	-	1	-	-	-	2	-	1	4	
PV-TF-M3S-14	3-1/2" x 14" Face, 1 Strut	599	1	-	-	1	-	-	-	2	-	1	-	-	-	2	-	1	4	
PV-TF-M3S-20	3-1/2" x 21" Face, 1 Strut	698	1	-	-	1	-	-	-	-	2	1	-	-	-	2	-	1	4	
PV-TF-M4S-8	4-1/2" x 8" Face, 1 Strut	559	1	-	-	1	-	-	-	-	-	1	2	-	-	-	2	-	4	
PV-TF-M4S-14	4-1/2" x 14" Face, 1 Strut	690	1	-	-	1	-	-	-	-	-	1	2	-	-	-	2	-	4	
PV-TF-M4S-20	4-1/2" x 21" Face, 1 Strut	841	1	-	-	1	-	-	-	-	-	1	-	2	-	-	2	-	4	
PV-TF-M#S2-#	MW Dish Face Mount with (2) Strut Support																			
PV-TF-M2S2-8	2-3/8" x 8" Face, 2 Strut	484	1	-	-	2	2	-	-	-	-	1	-	-	-	4	-	2	4	
PV-TF-M2S2-14	2-3/8" x 14" Face, 2 Strut	531	1	-	-	2	-	2	-	-	-	1	-	-	-	4	-	2	4	
PV-TF-M2S2-20	2-3/8" x 21" Face, 2 Strut	579	1	-	-	2	-	-	2	-	-	1	-	-	-	4	-	2	4	
PV-TF-M3S2-8	3-1/2" x 8" Face, 2 Strut	549	1	-	-	2	-	-	2	-	-	1	-	-	-	4	-	2	4	
PV-TF-M3S2-14	3-1/2" x 14" Face, 2 Strut	648	1	-	-	2	-	-	-	2	-	1	-	-	-	4	-	2	4	
PV-TF-M3S2-20	3-1/2" x 21" Face, 2 Strut	747	1	-	-	2	-	-	-	2	1	-	-	-	-	4	-	2	4	
PV-TF-M4S2-8	4-1/2" x 8" Face, 2 Strut	609	1	-	-	2	-	-	-	-	1	2	-	-	-	4	-	2	4	
PV-TF-M4S2-14	4-1/2" x 14" Face, 2 Strut	739	1	-	-	2	-	-	-	-	1	2	-	-	-	4	-	2	4	
PV-TF-M4S2-20	4-1/2" x 21" Face, 2 Strut	890	1	-	-	2	-	-	-	-	1	-	2	-	-	4	-	2	4	
PV-TF-ML#S-#	MW Dish Face Mount with (1) Strut Support, 14" Standoff																			
PV-TF-ML2S-8	2-3/8" x 8" Face, 1 Strut, 14" Standoff	470	1	-	-	1	2	-	-	-	-	1	-	-	-	2	-	1	4	
PV-TF-ML2S-14	2-3/8" x 14" Face, 1 Strut, 14" Standoff	518	1	-	-	1	-	2	-	-	-	1	-	-	-	2	-	1	4	
PV-TF-ML2S-20	2-3/8" x 21" Face, 1 Strut, 14" Standoff	565	1	-	-	1	-	-	2	-	-	1	-	-	-	2	-	1	4	
PV-TF-ML3S-8	3-1/2" x 8" Face, 1 Strut, 14" Standoff	534	1	-	-	1	-	-	2	-	-	1	-	-	-	2	-	1	4	
PV-TF-ML3S-14	3-1/2" x 14" Face, 1 Strut, 14" Standoff	633	1	-	-	1	-	-	-	2	-	1	-	-	-	2	-	1	4	
PV-TF-ML3S-20	3-1/2" x 21" Face, 1 Strut, 14" Standoff	732	1	-	-	1	-	-	-	-	2	1	-	-	-	2	-	1	4	
PV-TF-ML4S-8	4-1/2" x 8" Face, 1 Strut, 14" Standoff	587	1	-	-	1	-	-	-	-	-	1	2	-	-	-	2	-	4	
PV-TF-ML4S-14	4-1/2" x 14" Face, 1 Strut, 14" Standoff	717	1	-	-	1	-	-	-	-	-	1	-	2	-	-	2	-	4	
PV-TF-ML4S-20	4-1/2" x 21" Face, 1 Strut, 14" Standoff	868	1	-	-	1	-	-	-	-	-	1	-	2	-	-	2	-	4	
PV-TF-ML#S2-#	MW Dish Face Mount with (2) Strut Support, 14" Standoff																			
PV-TF-ML2S2-8	2-3/8" x 8" Face, 2 Strut, 14" Standoff	518	1	-	-	2	2	-	-	-	-	1	-	-	-	4	-	1	2	
PV-TF-ML2S2-14	2-3/8" x 14" Face, 2 Strut, 14" Standoff	566	1	-	-	2	-	2	-	-	-	1	-	-	-	4	-	1	2	
PV-TF-ML2S2-20	2-3/8" x 21" Face, 2 Strut, 14" Standoff	613	1	-	-	2	-	-	2	-	-	1	-	-	-	4	-	1	2	
PV-TF-ML3S2-8	3-1/2" x 8" Face, 2 Strut, 14" Standoff	583	1	-	-	2	-	-	-	2	-	1	-	-	-	4	-	1	2	
PV-TF-ML3S2-14	3-1/2" x 14" Face, 2 Strut, 14" Standoff	682	1	-	-	2	-	-	-	2	-	1	-	-	-	4	-	1	2	
PV-TF-ML3S2-20	3-1/2" x 21" Face, 2 Strut, 14" Standoff	781	1	-	-	2	-	-	-	-	2	1	-	-	-	4	-	1	2	
PV-TF-ML4S2-8	4-1/2" x 8" Face, 2 Strut, 14" Standoff	636	1	-	-	2	-	-	-	-	-	1	2	-	-	-	4	-	2	
PV-TF-ML4S2-14	4-1/2" x 14" Face, 2 Strut, 14" Standoff	766	1	-	-	2	-	-	-	-	-	1	2	-	-	-	4	-	2	
PV-TF-ML4S2-20	4-1/2" x 21" Face, 2 Strut, 14" Standoff	719	1	-	-	2	-	-	-	-	-	1	-	2	-	-	4	-	2	

2 OF 7

2/9/2021

SCALE 1:1

PERFECT VISION

01_Lattice Tower

06 Face Mounts

PV-TF-Tower Face Mount

1

2

3

4

1/28/21

11/27/19

1/13/16

2

DIMENSIONS ARE IN INCHES

TOLERANCES: ±.015"

ANGULAR PROFILE: ±1/16"

ALL OTHERS: ±1/16"

CHECKED: DJN

INIT: INT

STATUS: APPROVED

DATE: 1/28/21

REVISION: 2

DESCRIPTION: TOWER FACE MOUNT

DOCUMENT NUMBER: TF-ENG-01-R2

REV: 2

2 OF 7

2/9/2021



SCALE 1:1

01. Lattice Tower

06. Face Mounts

REPLACED BY PV-TF-ML-MB, PV-CO-U

REPLACED BY PV-SMART-MASK45

CHECKED: DUN

INIT

DATE: 11/21/19

STATUS: APPROVED

REV: 0

DATE: 1/31/16

PERFECT VISION

TOWER FACE MOUNT

TF-ENG-01-R2

2

REV

	
TOWER FACE MOUNT	
DOCUMENT NUMBER	REV
TF-ENG-01-R2	2